

Cost and Production Report

University of Twente

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1 Production

1.1 Introduction

SpeedMass is currently working on optimizing the layout of its toy car production line to reduce costs and improve efficiency.

As part of the project, the team made assumption-based calculations, conducted real-life testing, and developed a simulation model using "FlexSim" to evaluate and compare various production configurations.

Before diving into simulation details, key performance indicators (KPI's) were defined to guide the project and measure success across different scenarios. A leading KPI that affects both cost and production effectiveness is:

- **Time required to produce one unit:** reducing this time improves the performance and speed of producing subsequent cars, and consequently lowers overall production costs.

This indicator formed the basis for the evaluation of different production setups.

1.2 Cycle time analysis and workstation task allocation

In order to determine the most optimal layout for the production line, the takt time had to be calculated.

$$\text{Takt time} = \frac{\text{available production time}}{\text{customer demand}} = \frac{8 \cdot 229 \cdot 3600}{200,000} = 32.976 \text{ seconds/unit}$$

- 200,000 units are assumed to be sold.
- Based on average working hours in the Netherlands (where SC-04, our product, will be manufactured), it is assumed that the production line will operate 8 hours a day, 5 days a week. And, it is assumed that the worker has 20 paid vacation days and 11 national holidays (which in the worst-case scenario are all during the week) so in total, 229 workable days.
- At this stage, the possibility of having two shifts is not considered.

This calculation implies that no workstation should take more than 32.976 seconds per unit (toy car), as this would create a bottleneck in the production line. Since the longest process -for now, since is based on assumptions- is the injection molding (30 seconds) and is slightly under the takt time, there are no bottlenecks.

Following a thorough evaluation, the assembly part process was broken down into the following workstations (WS):

- **WS1:** Fixing two wheels to the shaft (press-fitted twice).
Estimated process time: 15 seconds
- **WS2:** Attaching the shafts (with wheels) to the frame and put on the bolts, nuts and washers.
Estimated process time: 20 seconds
- **WS3:** Assembling the frame (with shafts) to the bottom plate using bolts, nuts and washers.
Estimated process time: 20 seconds

- **WS4:** Snapping the body onto the frame and performing a quality check.
Estimated process time: 15 seconds

Based on the available information and assuming that both the vacuum forming and laser cutting processes each take 25 seconds, the overall cycle time is determined by the longest operation. In this case, the injection molding process is the limiting factor, with a duration of 30 seconds. Therefore, the cycle time is 30 seconds, meaning that one product (a single toy car unit) is completed every 30 seconds. However, the total processing time or first cycle duration time will be $30 + 15 + 20 + 20 + 15 = 100$ seconds.

Also, to meet production goals efficiently, the cycle time (how long it takes to produce one unit) should align with the takt time (the maximum time allowed per unit to meet customer demand). In this case, the demand can be met because the takt time (32.976 seconds) is slightly higher than the cycle time (30 seconds). However, it is important to note that the time that the processes take is still an estimate at this point of the project.

If we wanted to increase the gap between cycle time and takt time, one option could be to run the injection molding machine during the night. This would ensure that parts are pre-produced before regular working hours. But, since this process is not currently a bottleneck, there is no need to do so.

Moreover, reducing the cycle time even further could lead to overproduction, which goes against lean manufacturing principles.

On the other hand, there is a significant chance that some units may be defective. To account for this, producing slightly more than the required quantity —about 0.5% - 2.5% extra— could be a smart strategy. For example, if the original demand is 200,000 toy cars, this adjustment would raise the production target to 201,000 units and 205,000 units respectively, this can help ensure delivery even in case of defects.

$$\text{New takt time for 0.5\%} = \frac{\text{available production time}}{\text{customer demand}} = \frac{8 \cdot 229 \cdot 3600}{201,000} = 32.812 \text{ seconds/unit}$$

$$\text{New takt time for 2.5\%} = \frac{\text{available production time}}{\text{customer demand}} = \frac{8 \cdot 229 \cdot 3600}{205,000} = 32.172 \text{ seconds/unit}$$

1.3 Comparison to real-life simulation

After conducting a real-life simulation —where team members took on the role of assembly line workers— several new insights emerged. The most significant finding was that actual task durations differed from the initial estimates (see estimations in section 1.2).

For example, Workstation 4 (WS4) took significantly less time than expected, averaging only 5.90 seconds. The associated quality check also required minimal time. In contrast, Workstation 3 (WS3) became the slowest step in the process, with an average duration of 45.25 seconds—much higher than originally anticipated.

Based on these observations, the team redistributed tasks to better balance the workflow. Specifically, the final task was moved from WS4 to WS3, and half of WS3's original task was reassigned to WS2.

Additionally, a new workstation was introduced to address a quality issue: sharp edges on the wheels. This station, designated as Workstation 1, was responsible for cleaning all four wheels to ensure safety and consistency. The task averaged 21.6 seconds per unit and was deemed essential based on the condition of the wheels after manufacturing.

The injection molding process for the car frame was also performed in the real life. Due to the absence of ejector pins, the process took significantly longer than expected —approximately 94

seconds—. This delay is considered a solvable design issue that had not yet been addressed. As such, the originally estimated time of 30 seconds will still be used for planning purposes. It is important to note that without the inclusion of ejector pins, this station would become a major bottleneck in the production line. During the real-life manufacturing process, it became clear that the sprue left on the product needed to be removed. For that reason, this task will be included in the updated workstation layout, and one of the operators will be responsible for it.

The vacuum forming process was also carried out and revealed an unexpectedly high production time of 60 seconds, which deviated from initial estimates -affecting the cycle time and bottleneck consideration-. This result had a significant impact on overall production predictions, and the implications of this finding are discussed further in the section 1.6.

1.4 Assembly and production line adjustments

Workstation 2 had an average task time of 33.18 seconds, which exceeds the defined takt time. However, due to the efficiency of the worker at Workstation 1—who consistently completed their tasks ahead of schedule—support was occasionally provided to Workstation 2. This intermittent assistance helped reduce the effective processing time, maintaining the overall flow of the assembly line without causing delays. It was also observed that the operator at Workstation 2 was still in the learning phase. With practice, it is expected that the task time will decrease to approximately 26 seconds.

During the simulation, it was observed that wheel holes had to be manually drilled before entering the assembly line. This process took approximately 14 minutes for a batch of 16 wheels.

If a single drilling workstation were added in the production line, when processing 4 wheels at a time, the process would take:

$$\frac{14}{4} = 3.5 \text{ minutes}$$

This duration is relatively high for a single production step. An alternative solution would be to add four parallel drilling workstations, each handling one wheel at a time. In that case, the processing time per wheel becomes:

$$\frac{14}{16} = 0.875 \text{ minutes} = 52.5 \text{ seconds}$$

This configuration significantly reduces the time required per batch of 4 wheels but, it still creates a bottleneck.

Also, it is important to note that, before drilling the holes, it was necessary to remove the sprue and runner from the wheels, as they were initially produced using injection molding.

However, to streamline the process, it will be assumed that the wheels are delivered pre-drilled and free of sprue and runner (which the supplier would remove beforehand). As a result, this step is outsourced and excluded from the internal production line. Nonetheless, the edge-cleaning step remains necessary and is therefore included in the production line as an additional workstation.

The new proposed distribution is as follows:

- **WS1:** Cutting the sharp edges of the 4 wheels.
Estimated process time: 22 seconds. Material: wheels drilled
- **WS2:** Fixing two wheels to the shaft (press-fitted twice).
Estimated process time: 26 seconds. Material: shafts
- **WS3:** Removing the sprue from the injection molded frame. Attaching the shaft (with wheels) to the frame and putting and tightening one of the bolts (with nut and washer) to assemble the frame (with shafts) to the bottom plate.
Estimated process time: 23 seconds. Material: frame, bolt, nut, washer, bottom plate

- **WS4:** Putting and tightening the second bolt (with the nut and washer), and then snap-fitting the car body onto the rest.

Estimated process time: 25 seconds. Material: bolt, nut, washer, body of the car

The current layout of the production line (shown below in figure 1) reflects the most recent test results and assumptions. However, further simulations may lead to layout adjustments to achieve improved efficiency.

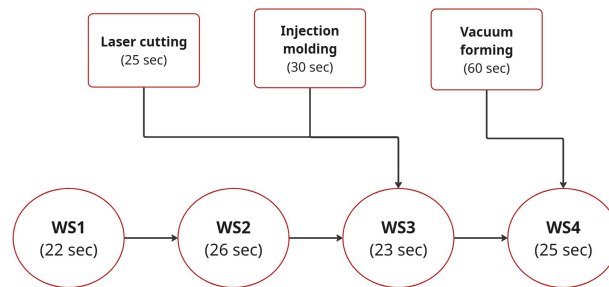


Figure 1: Mind map of current production line layout

1.5 Conclusion on the first production line layout

The production line underwent several iterations to better align with the initially defined takt time of 32.976 seconds per unit, based on a demand of 200,000 units annually. While the early estimates and task distributions provided a useful starting point, real-life simulations revealed discrepancies in actual task durations, identifying some inefficiencies and even forgotten steps.

Key insights led to a redistribution of tasks across workstations to balance workloads more evenly and reduce idle time. A new workstation was added to address sharp edges on the wheels—a critical quality concern not initially accounted for-. It was considered the fact that the wheels could come pre-drilled and with sprue and runner removed; and, a new task was taken into account, which is the removal of the sprue on the frames.

The layout, while already improved, remains flexible. Continued use of simulations like "FlexSim", combined with real-life testing will be key methods.

1.6 "FlexSim" simulations

In this section, the "FlexSim" program will be used to verify and evaluate the previous findings, with the goal of identifying the most optimal layout that can meet the demand in the shortest possible time (as defined by the leading KPI in Section 1.1).

This section is divided into three sub-sections: the first analyses the results of the current layout, the second explores potential improvements through further optimization, the third compares both configurations and the fourth adds everything else needed in the production line.

To achieve this, several factors will be considered:

- **Throughput:** how many products are completed
- **Average Cycle Time**
- **Queue length**

- **Operator Idle Time** (to reduce unnecessary labour cost):

$$\frac{\text{idle time}}{\text{total shift time}} \cdot 100 (\%)$$

- **Machine utilization** (check how efficiently machines are used):

$$\frac{\text{processing time}}{\text{available time}} \cdot 100 (\%)$$

1.6.1 Current layout

The first factor to be analyzed is whether the number of machines required (injection molding, laser cutting, and vacuum forming) is sufficient. For this reason, the initial simulation is kept simpler than the real-life setup. Additional processing steps and influencing factors will be introduced later in subsection 1.6.4. The "FlexSim" model will be built using the values obtained from the real-life simulation, as summarized in Figure 1.

The production line begins with three material sources (**Src.Body**, **Src.BottomPlate**, and **Src.Frame**) which feed into manufacturing machines: the Vacuum Former, Laser Cutter, and Injection Molding machine, respectively. These machines produce intermediate parts: the car body, bottom plate, and frame.

Parts are then queued and passed to a series of four workstations (the updated ones described on the Section 1.4).

Operators are assigned to each station, reflecting the human effort involved. Finished cars are collected at the FinishedCars output station.

- **Throughput**

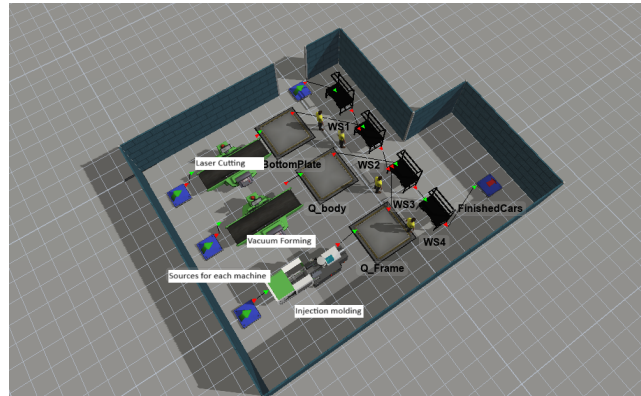


Figure 2: The current layout of the production line in "FlexSim"

Properties	
FinishedCars	
Statistics	
State	collecting
Throughput	
Input	Output
478.00	0.00

Figure 3: Throughput in 8 hours (1 working day)

With the current layout, the throughput is 478 units per day, which in 229 days (the assumed total working days in the worst-possible scenario), means:

$$478 \cdot 229 = 109,462$$

This output is below the market demand, indicating that a change in the production line is necessary. The main bottleneck was identified as the vacuum forming process, which was identified during the real-life testing. To address this issue, several solutions can be considered:

1. Add a faster cooling system to reduce cycle time
2. Operate the vacuum forming machine during night shifts
3. Add an additional vacuum forming machine to increase capacity

The first option, while potentially reducing cycle time, involves uncertain technical modifications and high upfront investment without guaranteed performance gains, making it an impractical short-term solution.

The second option, running the machine during night shifts, would increase output but at the cost of higher operational expenses, added staffing requirements, and logistical challenges—particularly in maintaining quality control and efficiency during extended hours.

Ultimately, the third option —adding a second vacuum forming machine— was selected. This approach directly and reliably increases capacity at the bottleneck without disrupting current operations. This option will be tested in the following subsection [1.6.2](#)

• Average Cycle Time

The average cycle time is obtained in the program FlexSim the following graph and average value:

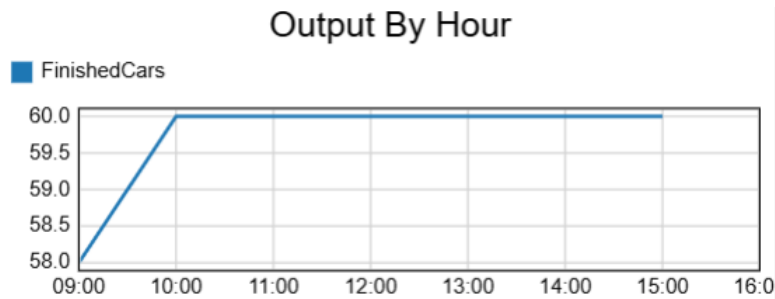


Figure 4: Output by hour in the current simulation

It is visible that at the very beginning there is some time of setup where the production is low, for the average cycle time that period will be ignored. Hence,

Object	Throughput
FinishedCars	59.97

Table 1: Average output/hour

With this collected value, the average cycle time can be calculated:

$$59.97 \frac{\text{unit}}{\text{hour}} \cdot \frac{1 \text{ h}}{3600 \text{ s}} = 0.0167 \frac{\text{unit}}{\text{s}} \Rightarrow \frac{1}{0.0167} = 60.024 \text{ seconds/unit}$$

The calculated average cycle time extracted from "FlexSim" gives 60.024 seconds/unit, which is approximately the double of the takt time. This means, that the demand is not met and therefore, optimization needs to be done.

So, after getting and evaluating the values of the throughput in 1 working day and the average cycle time for the production of one car, it can be concluded that the current layout needs to be changed somehow. In order to figure this out, although some solutions were proposed in the Throughput part; the queue length, the operator idle time and the machine utilization will be calculated in order to have the most reliable solution.

- **Queue length**

By using the 'Content' feature in FlexSim's Dashboard and selecting each queue individually, the following values were obtained from the simulation (table 2):

Object	Maximum	Average
Queue Bottom plate	635	313.87
Queue Body	1	0
Queue Frame	467	231.18

Table 2: Average and peak queue lengths during current simulation

The results clearly showed a large queue forming for the car body and frame, indicating delays in those areas. In contrast, the vacuum forming queue was consistently empty, confirming that vacuum forming is a bottleneck (as previously discussed in Section 1.3). Based on this information, it is now confirmed that the vacuum forming machine is the main issue and must be improved in some way.

- **Operator Idle Time**

To determine the workstations operators idle times—defined as the period when an operator is ready and waiting but cannot perform a task because the next one has not yet arrived—the "State" chart in the Dashboard is used (shown in figure 5)

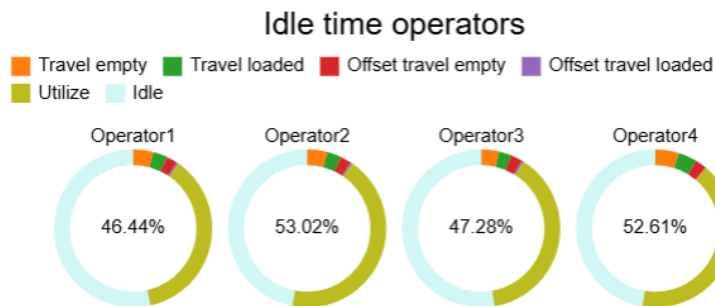


Figure 5: Productivity of operators pie chart from current "FlexSim"

The exact values of the operator idle times are the following:

Operator	Idle Time (%)
Operator 1	53.56%
Operator 2	46.98%
Operator 3	52.72%
Operator 4	47.39%

Table 3: Operator idle times

From Table 3, it can be observed that the operators have similar idle times, averaging around 50.16%. This means that, on average, operators are inactive for half of their working time, which leads to increased labor costs and reduced efficiency. Such a high idle time suggests that the issue may be linked to the bottleneck caused by the vacuum forming machine. Therefore, resolving the machine-related delays may also help reduce operator idle time. Additionally, in the optimized simulation, we will assume improved worker efficiency, including the possibility of cross-station support to further minimize idle periods.

- **Machine utilization**

To evaluate machine efficiency, the "State" chart in the Dashboard was used. The following pie chart was generated by the program to illustrate each machine's utilization:

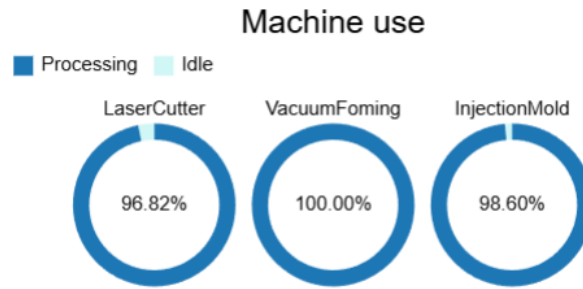


Figure 6: Machine utilization pie charts of current layout from FlexSim

The machine utilization chart shows that all three machines are operating at very high efficiency levels. The Vacuum Forming machine is at 100% utilization, confirming that it is fully occupied throughout the simulation and unable to keep up with demand. This reinforces its role as a clear bottleneck in the production line.

1.6.2 Improved layout

To meet the production demand, some of the results from Section 1.6.1 must be improved. As previously discussed, adding a second vacuum forming machine is a logical and necessary step to eliminate the current bottleneck. Additionally, while the previous simulation used the real-life simulation operator times, the new optimized simulation will assume that operators perform their tasks more quickly, as they become more experienced and familiar with the process.

This would result in the following mind map shown in figure 7:

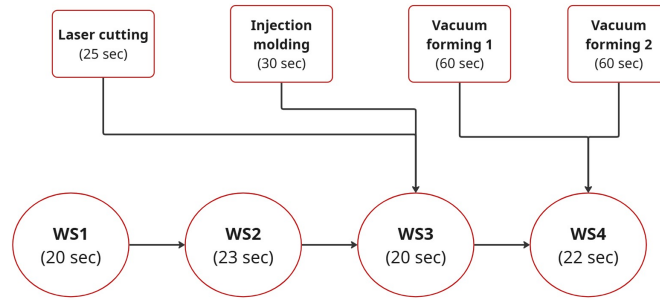


Figure 7: Improved layout of the production line

- **Throughput**

And in the FlexSim it would like like shown in figure 8

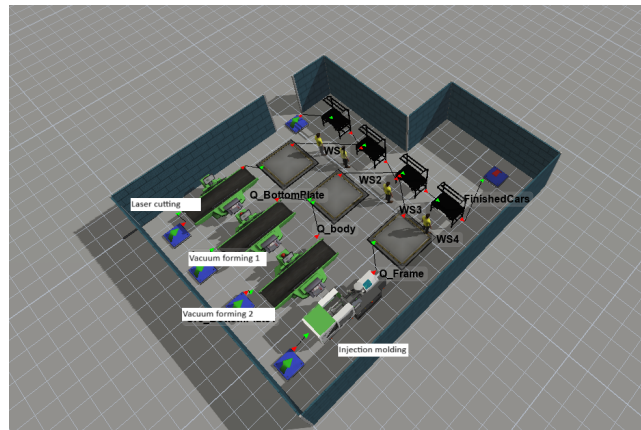


Figure 8: Updated layout of the production line

Now, the throughput is as shown below:

Properties	
FinishedCars	
Statistics	
State	collecting
Throughput	
Input	Output
944.00	0.00

Figure 9: Throughput in 8 hours (1 working day) of improved layout

With the updated layout, we obtain the following amount of toy cars after 209 days:

$$944 \cdot 229 = 216,176 \text{ units in 1 year.}$$

The annual market demand is 200,000 units. Based on the current workflow and simulation results, our production line is capable of delivering 216,176 units per year. This exceeds the

target by 16,176 units, which can serve as a buffer to cover potential defects, quality issues, or unexpected rework —ensuring the full demand can still be met reliably—.

- **Average Cycle Time**

The average cycle time is obtained in the program FlexSim the following graph and average value:

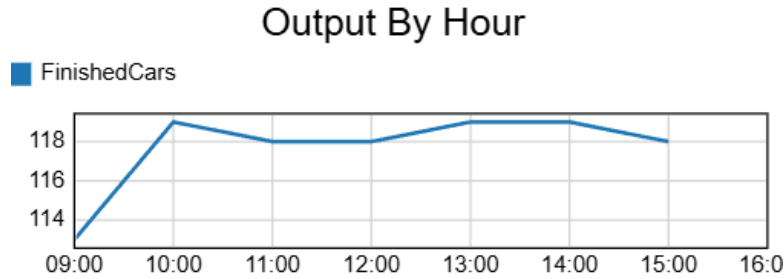


Figure 10: Output by hour in the improved simulation

It is visible that at the very beginning there is some time of setup where the production is low, for the average cycle time that period will be ignored. Hence,

Object	Throughput
FinishedCars	118

Table 4: Average output/hour

With this collected value, the average cycle time can be calculated:

$$118 \frac{\text{unit}}{\text{hour}} \cdot \frac{1 \text{ h}}{3600 \text{ s}} = 0.033 \frac{\text{unit}}{\text{s}} \Rightarrow \frac{1}{0.033} = 30.30 \text{ seconds/unit}$$

The average cycle time is under the takt time of 32.976 which is coherent since that means the demand is met and it was seen in the Troughput part that that is true (216,176 units will be produced) This confirms that, under the simulated conditions, the system is capable of meeting and slightly exceeding the market demand.

However, even if the target values are achieved, it is important to check for potential overproduction. One machine may be producing more than the others, leading to imbalances. For this reason, queue lengths will be monitored and taken into account.

- **Queue length**

The queue lengths of the improved layout is shown below in 7:

Object	Maximum	Average
Queue Bottom plate	169	82.58
Queue Body	13	7.34
Queue Frame	2	0.05

Table 5: Average and peak queue lengths during improved simulation

The table shows that the bottom plates have the highest average queue length, with 82.58 units. In contrast, the other components do not exhibit any significant queue buildup. In the section 1.6.3 the values will be compared with the old layout ones.

- **Operator Idle Time**

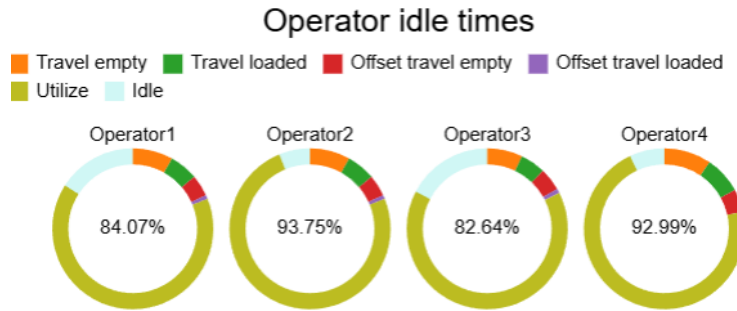


Figure 11: Productivity of operators pie chart from FlexSim

The exact values of the operator idle times are the following:

Operator	Idle Time (%)
Operator 1	15.93%
Operator 2	6.25%
Operator 3	17.36%
Operator 4	7.01%

Table 6: Operator idle times

From the data, it is evident that Operator 1 and Operator 3 have the highest idle times —approximately 17.07% and 10.39%, respectively—. These values indicate significant periods of unproductive waiting, likely due to imbalances in task allocation or delays in upstream processes.

To reduce idle time and improve overall labor efficiency, cross-support between operators could be implemented—particularly by enabling Operator 1 and Operator 3 to assist other stations during their idle periods. In fact, during the real-life simulation, this strategy was applied informally: the worker at WS1 assisted WS2, and the operator at WS3 supported WS4 whenever possible.

- **Machine utilization**

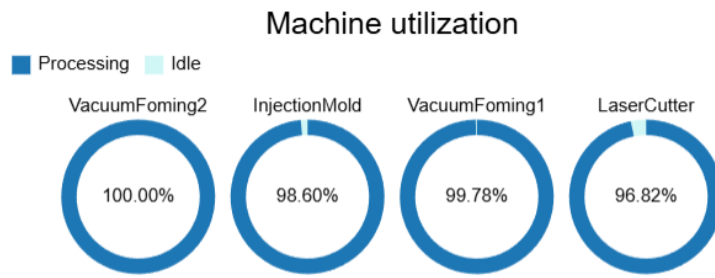


Figure 12: Machine Utilization pie charts from FlexSim

The machine utilization values indicate that all machines are being used highly efficiently, with utilization rates above 96%. In particular, VacuumFoming2 is operating at 100% utilization, suggesting that while the system is currently well-balanced, this machine may become a bottleneck under higher demand or during unexpected downtime.

1.6.3 Comparison

First, the throughput shows a clear difference between the two simulations. The first setup produced 109,462 units per year, which is below the required demand. In contrast, the second setup achieved 216,176 units per year, fully meeting—and even exceeding—the target. This confirms that the second simulation represents a clear improvement over the first.

Then, the output by hour is in consequence different as well. Below in figure 13 the graphs shown in figure 4 and 10 are put together.

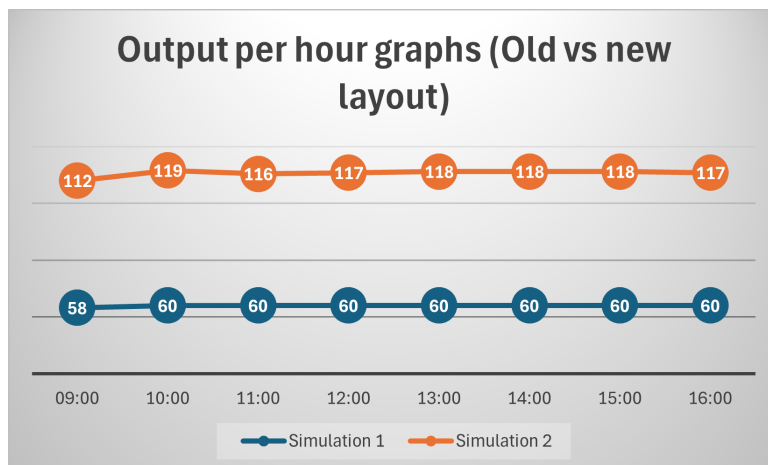


Figure 13: Machine Utilization pie charts from FlexSim

It can be clearly seen that the average cycle time of the second simulation is higher which makes sense since the throughput is also higher.

For the queue length comparisons the average values will be taken

Object	Average (Simulation 1)	Average (Simulation 2)
Queue Bottom plate	313.87	82.58
Queue Body	0	7.34
Queue Frame	213.18	0.05

Table 7: Average and peak queue lengths during improved simulation

The average queue lengths show a significant improvement in the optimized simulation (Simulation 2) compared to the original configuration (Simulation 1).

1. The bottom plate queue dropped from 313.87 to 82.58, indicating that the excessive buildup of parts was successfully reduced.
2. The frame queue was reduced drastically from 213.18 to just 0.05, almost eliminating idle inventory in that stage.
3. Although the body queue increased slightly from 0 to 7.34, this value is still very low and not critical.

Overall, the optimization not only improved throughput but also reduced inventory accumulation, which contributes to a more efficient production flow.

For the idles times a graph has been done to visually see the difference (figure 14)

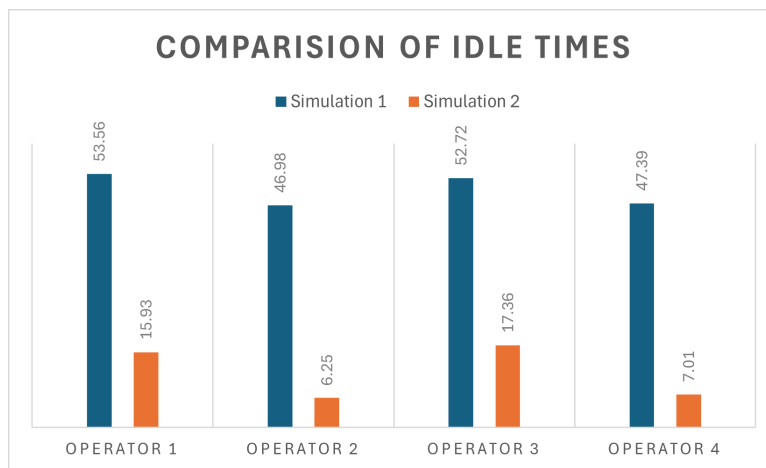


Figure 14: Idle times comparison

There is a significant difference in idle times between Simulation 1 and Simulation 2, with Simulation 2 showing much lower idle times.

Lastly, machine utilization is very high in both simulations, with values that are very similar. For this reason, a detailed analysis is not necessary.

1.6.4 New manufacturing steps included in the production line

At this point, there is an effective number of machines to produce the parts and a well-distributed workstation system to assemble the toy car. However, some production steps were not previously considered, and the sources for different materials at each workstation were also not included in the FlexSim simulation—even though they were part of the original plan—:

1. Drilling holes in the propylene wheels

Since the toy car has shafts attached to the wheels, holes must be drilled. This step was originally assumed to be outsourced (see Section 1.4). Nevertheless, a simulation was carried out with the corresponding processing time to validate the impact on the production line. In this scenario, four workstations were added, each with a processing time of 52.5 seconds.

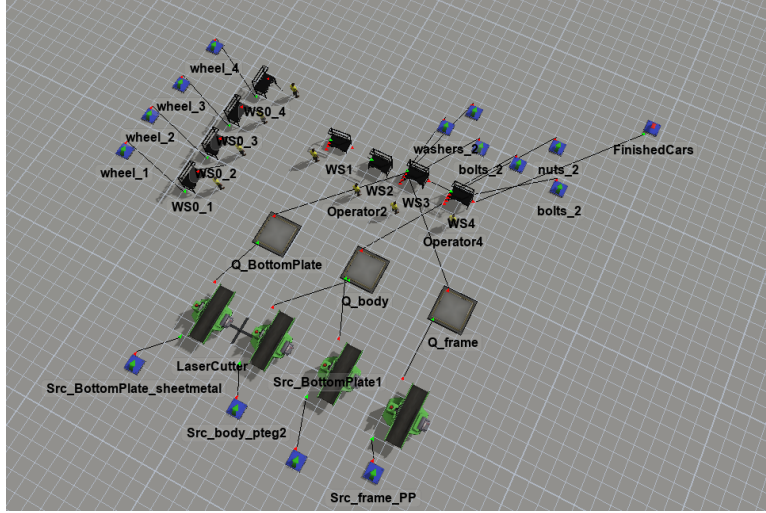


Figure 15: Improved layout with wheel drilling workstations (x4)

With this simulation, the target throughput could not be reached. The output achieved was 478 units, while 873.36 units are required to meet the demand exactly. Hence, outsourcing is still the best option.

2. Cutting the shafts

The company SpeedMass, which produces the toy car model SC-04, supplies 6-meter shafts. However, each toy car only requires shafts of 81.5 mm. Therefore, a workstation equipped with a hand saw should be added to cut the shafts to size. It is assumed that this process takes 10 seconds per car (5 seconds per shaft, as two shafts are needed per unit) and then it is transported to the next step.

3. Cutting the polypropylene sheets for vacuum forming

SpeedMass supplies polypropylene (PP) sheets measuring 1000×1000 mm and 2 mm thick. These sheets must be cut into 30×30 mm squares for production. To perform this step, a workstation with a die-cutting machine operated manually is proposed. It is assumed that each cut takes approximately 10 seconds, accounting for the need for precise measurements.

4. Trimming the edges of the polypropylene sheet after vacuum forming

To remove excess material and ensure clean edges after vacuum forming, we should use a machine like CNC router. This tool provides high precision and is well-suited for the complex shapes of the car body. In the real-life simulation, the trimming was performed manually, resulting in an average processing time of 40 seconds per part—a duration considered too high for efficient production—so, it will be assumed that our CNC router cuts them in 15 seconds.

5. Packaging

Packaging will be outsourced and therefore excluded from the production line. The reasoning behind this is explained in section 2.3.3 as it is primarily a cost-based decision.

After introducing these new steps, a new simulation will be conducted to verify whether the production line still meets the required conditions. First, we will simulate the following layout:

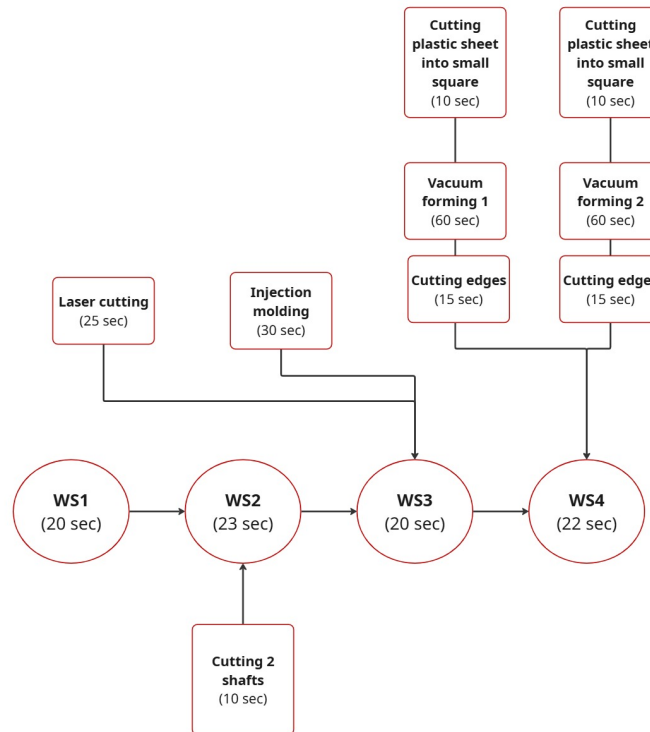


Figure 16: Production line with all steps included

Which in the FlexSim looks something like the following:

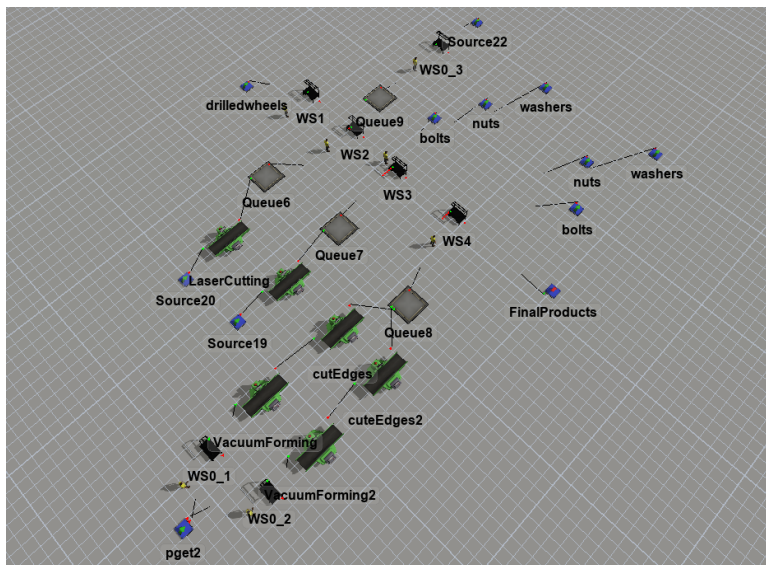


Figure 17: Production line with all steps included in FlexSim

With this configuration, the production line achieves a throughput of 895 toy cars per day, which

is sufficient to meet the demand. Therefore, layout modifications are not enteriley necessary. If there would be necessary, one potential solution would be to outsource the step preceding the vacuum forming process. This scenario was simulated, and the results showed that outsourcing this step had no impact on throughput. As such, the decision to keep or remove the related machine should depend primarily on cost considerations—specifically, whether outsourcing is more cost-effective (for now, these workstations will be retained, as their presence does not affect the production output according to the simulation data).

To explore potential optimizations related to operator efficiency, a similar approach to the one used in Sections 1.6.1 and 1.6.2 will be applied:

- **Queue length**

Object	Maximum	Average
Queue Bottom plate	335	167.88
Queue Body	182	91.66
Queue Frame	341	170.79
Queue Shafts	176	87.47

Table 8: Average and peak queue lengths during new simulation

As shown in table 8, the queue lengths are higher than in the previously improved layout (see subsection 1.6.2). This increase is due to the addition of new steps that slow down the overall process. To address this, a potential solution involves assigning the operator at WS0 (responsible for cutting the shafts) an additional role. Since this step is relatively quick and occurs at the beginning of the production flow, the operator can pause after cutting twice the required throughput—since they have to cut 2 shafts—. They could then take on a secondary role in machine maintenance, such as stopping or assisting with machines when needed. This way, their workstation (WS0 in FlexSim) wouldn't have any queue and the other ones could be minimized.

- **Operator Idle Time** (to reduce unnecessary labour cost):

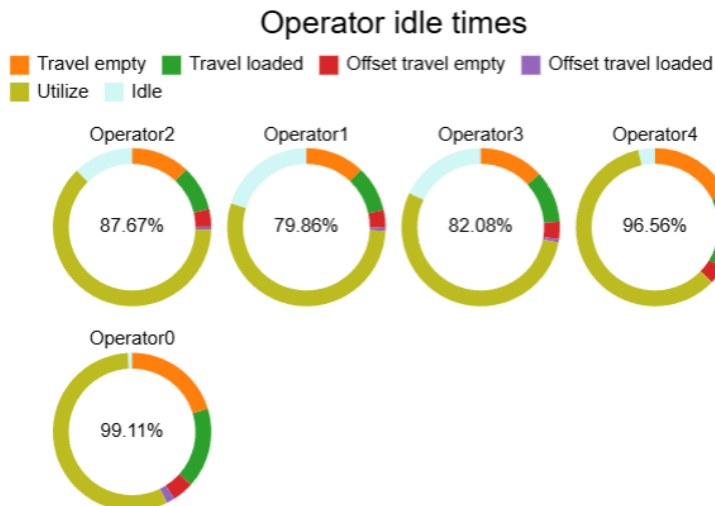


Figure 18: Efficiency of workers in the new layout (FlexSim simulation)

All operators are currently working efficiently, and overall efficiency is expected to improve further once Operator 0 (assigned to WS0) takes on the additional role of monitoring and

assisting with machine operations.

Machine utilization is expected to be high and does not require separate analysis. Similarly, reporting the average cycle time is unnecessary, since the overall throughput has already been determined.

Therefore, the production line layout shown in Figure 16 is the most efficient and optimized configuration that meets the targeted demand, and will be adopted by the company SpeedMass.

1.7 Storage evaluation

It was decided to store raw materials within the factory premises, as this approach offers several operational and financial advantages aligned with cost optimization.

For SpeedMass, immediate accessibility to raw materials is crucial to ensure that production remains within the time constraints defined by the takt time. As such, materials will be stored on-site in the main inventory area, rather than in a separate location with additional utility and rental costs. This proximity significantly reduces lead time between material retrieval and production, enabling a smoother workflow and minimizing delays.

If off-site storage were chosen, several challenges would arise, including:

- Additional transportation costs
- Time delays due to transit and potential traffic issues
- Increased logistical complexity
- Greater risk of production disruption

By eliminating the need for third-party warehousing or external storage, SpeedMass reduces fuel consumption, minimizes handling delays, and improves production reliability.

Moreover, on-site storage enhances coordination between the inventory and production departments. It enables real-time inventory tracking, more accurate stock management, reduced overstocking, and fewer interruptions due to material shortages.

1.8 Inventory

At this point, a finalized layout has been established, but one key concept remains to be addressed: inventory. It is essential to consider the inventory required to meet the production goals efficiently.

Given the new throughput of 895 toy cars per day and assuming 229 working days in a year, the total annual production is:

$$229 \cdot 895 = 204,955 \text{ toy cars in 1 year}$$

This output satisfies the demand but results in a buffer of 4,955 units, representing approximately 2.5% more than the required production. However, the company had initially set a target buffer of 0.5-2.5% so, it is inside the range. Therefore, the value provided by the simulation (895 units/day) will be used as the reference for production planning.

To ensure uninterrupted production at this throughput level, it is necessary to calculate the required inventory for each raw material used in the manufacturing process. These materials, supplied by The Cube, include:

- 3-meter M4 steel shafts
- Pre-drilled wheels

- M4 Bolts, M4 washers and M4 nuts
- Sheet metal for bottom plates
- PETG sheet for frames
- Polypropylene (PP) sheets for vacuum forming

The inventory quantities will be calculated based on the total annual production volume of 204,955 toy cars, while considering the specific consumption rate of each item per unit produced.

1.8.1 Inventory calculation per material

1. 3-meter M4 steel shafts

$$\frac{3000 \text{ mm}}{81.5 \text{ mm}} \approx 36.81 \Rightarrow 36 \text{ usable pieces per shaft}$$

$$\frac{36}{2} = 18 \text{ cars produced with one 3-meter shaft}$$

$$\frac{204,955}{18} \approx 11,386 \text{ 3-meter bars}$$

Required inventory: *11,386 steel shafts of 3 m.*

2. Pre-drilled wheels

Each toy car requires 4 wheels:

$$204,955 \cdot 4 = 819,820 \text{ wheels}$$

Required inventory: *819,820 pre-drilled wheels.*

3. M4 bolts, M4 washers, and M4 nuts

Each toy car requires 2 of each component (bolts, washers, and nuts). For an annual production of 204,955 toy cars, the required quantity of each component is:

$$204,955 \cdot 2 = 409,910 \quad (\text{for each: bolts, washers, and nuts})$$

Required inventory: *409,910 M4 bolts, 409,910 M4 washers, and 409,910 M4 nuts.*

4. Sheet metal for bottom plates

To determine the number of bottom plates that can be cut from each standard metal sheet (2,000 mm × 1,000 mm), two methods were considered: a theoretical area-based approach and a practical fit-by-dimension approach.

The first method divides the total surface area of the sheet by the area of a single bottom plate:

- Sheet area: $2,000 \times 1,000 = 2,000,000 \text{ mm}^2$
- Bottom plate area: $121 \times 56 = 6,776 \text{ mm}^2$
- Theoretical maximum: $2,000,000 \div 6,776 \approx 295$ bottom plates per sheet

This estimate assumes a perfect arrangement with no wasted space, which is rarely achievable in practice due to geometric constraints and required spacing between cuts.

In real-world manufacturing, cutting is constrained by the dimensions of the sheet and the part. Therefore, the number of bottom plates is calculated by fitting full rows and columns along the sheet's edges:

- Along the 2,000 mm side: $\left\lfloor \frac{2,000}{121} \right\rfloor = 16$ plates
- Along the 1,000 mm side: $\left\lfloor \frac{1,000}{56} \right\rfloor = 17$ plates
- Total usable plates per sheet: $16 \times 17 = 272$ bottom plates

This method provides a more realistic and reliable result for production planning, as it reflects how parts are actually arranged and cut.

The practical fit-by-dimension value of 272 plates per sheet is used for inventory calculations. For an annual production of 204,955 toy cars, the number of sheets required is:

$$\frac{204,955}{272} \approx 754 \text{ metal sheets}$$

Required inventory: *754 metal sheets*

5. PETG sheets for frames

The vacuum-formed body of each toy car is made from a 30 mm $\times$ 30 mm square cut from a standard 1,000 mm $\times$ 1,000 mm polypropylene (PP) sheet.

- Sheet area: $1,000 \times 1,000 = 1,000,000 \text{ mm}^2$
- Part area: $30 \times 30 = 900 \text{ mm}^2$
- Theoretical maximum: $1,000,000 \div 900 \approx 1,111$ parts per sheet

This calculation assumes perfect space utilization without gaps, which is unrealistic for physical cutting.

To reflect how parts are actually cut in manufacturing, we calculate how many full 30 mm $\times$ 30 mm squares can fit across the sheet:

- Along the 1,000 mm side: $\left\lfloor \frac{1,000}{30} \right\rfloor = 33$ parts
- Along the other 1,000 mm side: $\left\lfloor \frac{1,000}{30} \right\rfloor = 33$ parts
- Total usable parts per sheet: $33 \times 33 = 1,089$ parts

The practical value of 1,089 usable parts per sheet is used for inventory planning. To meet the annual production of 204,955 toy cars:

$$\frac{204,955}{1,089} \approx 188 \text{ PP sheets}$$

Required inventory: *188 polypropylene sheets (1,000 mm $\times$ 1,000 mm)*

6. Polypropylene (PP) sheets for vacuum forming

Each PP sheet produces 900 units:

$$\frac{204,955}{900} \approx 228 \text{ sheets}$$

Required inventory: *228 polypropylene sheets.*

1.8.2 Summary of inventory amounts

Material	Required quantity
Steel shafts (3 m)	11,386
Pre-drilled wheels	819,820
Bolts	409,910
Nuts	409,910
Washers	409,910
Sheet metal (bottom plates)	754
PTEG sheets (frames)	188
PP sheets (vacuum forming)	228

Table 9: Total raw material inventory needed for annual production

1.9 Conclusion

The initial key performance indicator (KPI) defined at the beginning of the project was the time required to produce one unit, which directly influences production efficiency and cost. Through a structured methodology—starting with theoretical estimations, followed by real-life testing, and finalized through simulation using FlexSim—this KPI was successfully addressed and optimized.

The first simulation revealed several inefficiencies, most notably a bottleneck at the vacuum forming stage and high idle times for operators. These findings guided a series of targeted improvements: tasks were redistributed among workstations to balance workloads, operator efficiency was improved based on learning effects, and a second vacuum forming machine was added to eliminate the bottleneck. The second simulation, which included the addition of a new vacuum forming machine, showed significant improvement in different factors like throughput, operator idle time and queue length. However, it did not account for several essential production steps, for this reason, a third simulation was conducted. In this version, additional processing steps were included, along with new material sources that directly impact the assembly workstations. This simulation also introduced new considerations, such as the advantage of outsourcing the toy car wheels to receive them pre-drilled, the required inventory amounts for each material source, the appropriate storage locations for these raw materials, and alternative strategies to improve operator efficiency.

The final layout (see figure 16) includes:

- Four balanced workstations with reduced task times and minimal idle periods.
- Two vacuum forming machines to ensure continuous supply of car bodies.
- Parallel workstations to assure production flow
- One injection molding machine and one laser cutting machine
- One operator with a dual role to manage the queues of the frame, bottom plate, and body of the toy car
- Designated storage areas for raw materials, with quantities accurately accounted for

With a complete and realistic production layout in place—validated through progressive simulations—SpeedMass is now equipped with a manufacturing system that meets its production goals reliably, cost-effectively, and with future scalability in mind.

2 Costs

2.1 Introduction

A cost analysis for mass production is crucial, which will be made in the project in order to help the company to study the costs and expenses of the parts for the desired number of products consisting of different parts, like subassemblies such as wheels, shafts, and the bodies.

It serves as a guidance for the project to identify in a short term the areas and the steps, whereas expenses can be minimized without affecting the quality of product and, on a long term, the sales trend of Speed Mass. A closer look at the market and mass-produced toy cars helps in setting competitive prices by finding a balance between appropriate pricing and market competitiveness. By looking into the ordered parts prices and the needed processes with what materials will be used, cost analysis will navigate if there are any shortages, figure out the unnecessary components that would increase the efficiency of the mass production.

2.2 Calculations

The costs calculations can be divided into two main parts: machine costs and material costs for ordered parts. For ordered parts' cost, firstly the total cost for one car was calculated. For the machine costs, cost of each machine was calculated, which are injection molding, vacuum forming, laser cutting and CNC router (which was first introduced in the section 1.6.4. Also for the machines, in the end each cost was multiplied with the production quantity. After multiplying everything with the production quantity, assembly and packaging costs were also added. All the costs were summed up to find the mass production cost price and then the sum was divided to find the cost for one single car. All these steps will be shown in more detail, within calculations, in the following sections. Also in the appendix, each calculation can be seen.

2.2.1 Materials' costs for ordered parts

In this section only the cost of ordered parts will be explained. Materials used for the parts that were also used in the machines (like the frame that was made in injection moulding, bottom plate that was made with laser cutting, etc.) were calculated under machine prices. Ordered parts for the toy cars are wheels (with product number CB30E in Cube production, made from polypropylene), shafts (with 4 milimeter diameter and product number CBS4 in Cube production, made from steel), M5 nuts, M5 bolts, M5 washers, and ejector pins.

For finding the cost of ordered products per car, costs were used obtained from the price list in Cube Production. Each cost amount for given amount was divided by the amount of parts it corresponds to, and then multiplied by the amount it is present in a single car. Calculations are a bit different for the shafts, but it is explained below. Calculations can be seen for each part below:

$$\text{wheels: } \frac{5,85 \text{ €}}{100 \text{ pieces}} \cdot 4 = \mathbf{0,2340 \text{ €}}$$

$$\text{Shafts: } 6 \text{ meters of shaft} = 3,55 \text{ €} \implies 1 \text{ meter of the shaft} = 0,5917 \text{ €}$$

$$1 \text{ shaft} = 81,5 \text{ mm} = 0,0815 \text{ m}$$

$$\text{Shaft cost per car} = 0,0815 \cdot 0,5917 \text{ €} \cdot 2 = \mathbf{0,0964 \text{ €}}$$

$$\text{M5 nuts: } \frac{3,95 \text{ €}}{100 \text{ pieces}} \cdot 2 \text{ (pieces per car)} = \mathbf{0,0790 \text{ €}}$$

$$\text{M5 bolts: } \frac{6,99 \text{ €}}{150 \text{ pieces}} \cdot 2 \text{ (pieces per car)} = \mathbf{0,0932 \text{ €}}$$

$$\text{M5 washers: } \frac{9,85 \text{ €}}{500 \text{ pieces}} \cdot 2 \text{ (pieces per car)} = \mathbf{0,0394 \text{ €}}$$

$$\text{Ejector pins: } \frac{4,95 \text{ €}}{15 \text{ pieces}} \cdot 3 \text{ (pieces per car)} = \mathbf{0,9900 \text{ €}}$$

Apart from these ordered parts, outsourced drilled wheels are calculated in this section as well, which was mentioned in the section 1.4. The outsourcing price was estimated from the labour cost mainly. A market research was done for the drilling machines, and it was found out that compared to the other machines that were used in the SpeedMass factory, it had a much lower cost; and considering that it is a simpler machine the purchase price of the drilling machine will not have a big impact in the cost of drilled machines. Thus the labour cost was taken into account while estimating the outsourcing cost.

Time to drill 1 wheels was estimated to be 45 seconds. In the section 1.4 it was stated that it took around 54 seconds to drill one car, and considering that more experienced workers will be drilling this time was lowered to 45 seconds for the estimation. Thus, total time spent to drill one car's wheels (4 wheels) is 180 seconds. Consequently, the total time to drill every cars' wheels is 1000 hours. Within this information, the total price of drilling can be found by multiplying the total time to drill with employee cost per hour (25,1987 €) (the way employee cost per hour was found will be calculated and explained in the section 2.2.3). This calculation basically intends to find the labour cost that will be spend for the drilling. After the multiplication, the total price for milling was found to be 251986,90 €. Then, to find the drilling cost per car 27998,54 € was divided to the amount of toy cars, in other words to 200000. Drilling cost per car was found to be 1,26 €. This value will be added to the ordered parts per car.

Overall ordered parts cost for one car is:

$$\mathbf{0,2340 + 0,0964 + 0,0790 + 0,0932 + 0,0394 + 0,9900 + 0,14 = 1,6720 \text{ €}}$$

In appendix E these calculations can be seen in the excel sheet. Now that the cost of ordered parts's of one single car is known, cost for the mass production can be calculated by multiplying it with the production amount, as seen below:

$$1,6720 \text{ €} \cdot 200000 = \mathbf{334406,8777 \text{ €}}$$

2.2.2 Machines

Each machine has two parts for the machine cost: setup cost and the machine cost per product. The setup cost was calculated for all machines separately, and then added together as well as other components that will be shown. Machine costs were also calculated separately, and then used during the calculation of the final mass production cost. Firstly, the set up costs of every machine will be calculated, and then cost of each machine will be calculated. Before diving into calculations, the overall costs that will be calculated can be seen in the table below:

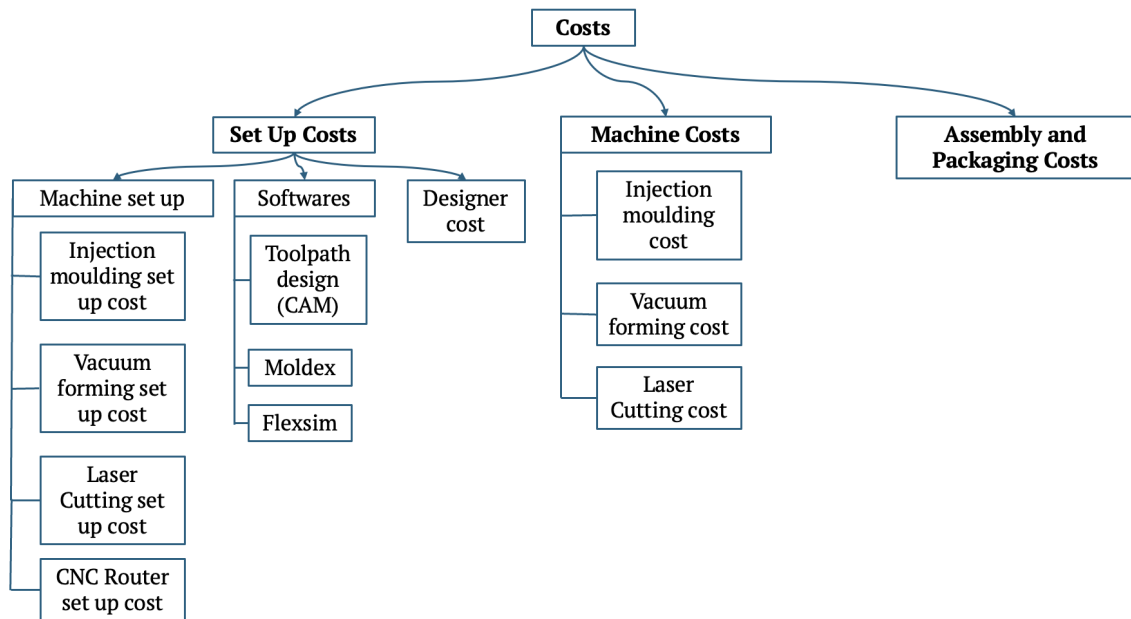


Figure 19: Costs Diagram

2.2.3 Set Up Costs

Set up cost of each machine includes depreciation cost, labour cost and tool cost, and set up time. Each of them is calculated for each machine.

Depreciation cost

Depreciation cost of the machine stands for the reduction cost after the machine life is complete, which refers to the decreased value of a machine over time as it is getting older. It was calculated by the formula below:

$$\text{Depreciation per hour} = \frac{\text{Purchase price of the machine (€)}}{\text{Estimated lifespan of the machine (year)} \cdot \text{Working hours (hours)}}$$

For each machine, the depreciation cost can be seen below:

$$\text{Injection moulding depreciation cost: } \frac{10000 \text{ (€)}}{10 \cdot 1832} = 0,5459 \text{ €}$$

$$\text{Vacuum forming depreciation cost: } \frac{20000 \text{ (€)}}{10 \cdot 1832} = 1,0917 \text{ €}$$

$$\text{Laser cutting depreciation cost: } \frac{138000 \text{ (€)}}{10 \cdot 1832} = 7,5328 \text{ €}$$

$$\text{CNC router depreciation cost: } \frac{50000 \text{ (€)}}{10 \cdot 1832} = 2,7293 \text{ €}$$

For the purchase price of the machines, they were determined from the models in the UT workshop. For injection molding, the machine BOY 22 A PRO was used, for vacuum forming Formech 450DT vacuum former was used, and for laser cutting Bodor i7 fiber laser cutting machine was used.

For working hours; because the production will be made in the Netherlands, working hours of the Netherlands were used. It was considered that each employee would work 8 hours a day. There are 52 weeks in a year, and 5 working days each week. So this makes 260 working days. From the Dutch government's website it was calculated that for an employee working 40 hours a week they have 160 hours of leave per year, and additionally there are 11 public holidays as well. This makes 31 days of holidays per year. After subtracting all the holidays, 229 working days remain. Considering that 1 working day is worth of 8 hours of working, this makes a total of 1832 working hours per year. These calculations were done by assuming the case with the most amount of holidays, which is the case when neither of the public holidays collide with the weekends (as well as the statutory leave) (Holiday Entitlement, 2020). Also for the employee cost per hour, average monthly salary in the Netherlands of an employee in industry was taken as a reference, which is 3847 € (Carter, 2017). And the estimated lifespan of the machine was estimated to be 10 years, after discussing with the Cube.

Labour Cost

After the depreciation cost, labour cost per hour calculation was done. Labour cost analysis is crucial because it helps to identify which areas need to be optimized. Since it is relevant to the production line, it opens the door for many observations such as analyzing how much time is spent on bottlenecks in the production lines, and opportunities for improvement. Labour cost corresponds to employee cost per hour times number of employees. The number of employees is determined as 1 for injection moulding, laser cutting and CNC router, and 2 for vacuum forming, the reason of choosing to have 2 vacuum forming machines instead of 1 was explained in the section 1.6.1. Calculation of employee cost per hour can be seen below:

$$\text{Employee cost per hour (€/hour)} : \frac{3847 \text{ €} \cdot 12}{1832 \text{ hours}} = \mathbf{25,1987 \text{ (€/hour)}}$$

Here, the monthly salary of an employee was multiplied by 12 months to find the salary of an employee in a year. Then, it was divided by the working hours per year to find the employee cost per hour. Using the employee cost, labour cost can be calculated:

Labour cost per hour (€/hour) for vacuum forming :

$$25,1987 \text{ (€/hour)} \cdot 2 \text{ (number of employees)} = \mathbf{50,3974 \text{ (€/hour)}}$$

Labour cost per hour (€/hour) for injection moulding, laser cutting and CNC router :

$$25,1987 \text{ (€/hour)} \cdot 1 \text{ (number of employees)} = \mathbf{25,1987 \text{ (€/hour)}}$$

Lastly, the tool cost is assumed to be 100€ for vacuum forming and laser cutting, 50€ for injection molding. Combination of all the calculations that has been done for each machine can be summed up with the calculations below:

Injection molding set up cost: (Depreciation cost + Labour cost) · Set up time + Tool Cost

$$= (0,5459 + 25,1987) \cdot 1 + 50 = 75,7445 \text{ €}$$

Vacuum forming set up cost: (Depreciation cost + Labour cost) · Set up time + Tool Cost

$$= (1,0917 + 50,3974) \cdot 1 + 100 = 151,4891 \text{ €}$$

Laser cutting set up cost: (Depreciation cost + Labour cost) · Set up time + Tool Cost

$$= (7,5328 + 25,1987) \cdot 1 + 100 = 132,7314 \text{ €}$$

CNC router set up cost: (Depreciation cost + Labour cost) · Set up time + Tool Cost

$$= (2,7293 + 25,1987) \cdot 1 + 100 = 127,9279 \text{ €}$$

Total Set Up Cost

After calculation of each machine's set up cost, there are three more components used during the calculation of set-up costs: designer cost, software costs (CAM, FlexSim, Moldex) and the hand saw cost (that is used to cut the shafts, which is a was mentioned in the section 1.6.4. For the hand saw cost, a market research was done and the average price was found to be 20€. For the designer cost, it was assumed that the designer worked for 2 days straight and works for 8 hours a day. Hourly salary of the designer (who is a mid level) is assumed as 30 €, based on research (Product Designer Salary in Netherlands, 2025). From that, the salary of the designer is 480 €. Overall set up cost calculation can be seen below:

Injection molding set up: 75,7445 €

Vacuum forming set up: 151,4891 €

Laser cutting set up: 132,7314 €

CNC router set up: 127,9279 €

Toolpath Design (CAM): 500 €

FlexSim: 500 €

Moldex: 500 €

Designer cost: 480 €

Hand saw cost: 20 €

Final set up cost = 2359,9651 €

2.2.4 Injection Moulding Machine Cost

The injection moulding machine cost will be calculated by multiplying injection moulding machine cost per product with the production quantity, which is 200000. Injection molding machine cost per product is calculated by the material cost, primary machine cost, secondary machine cost and overhead cost all added to each other. Overall components of the injection moulding machine cost per hour can be seen in the figure below:

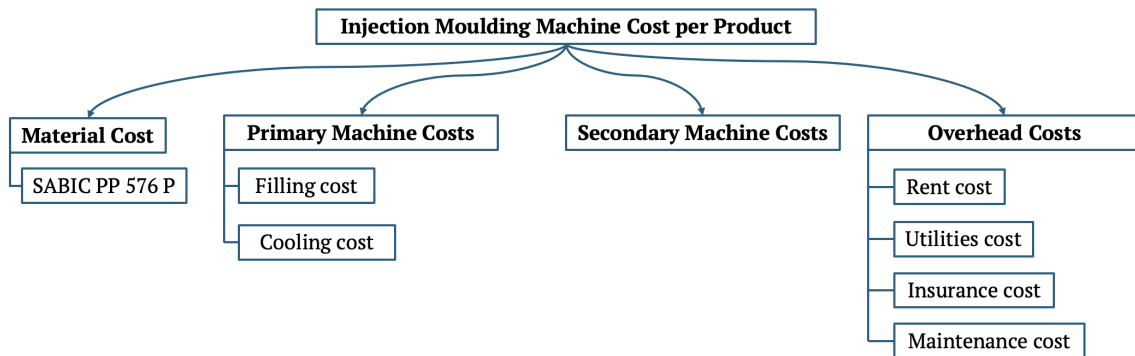


Figure 20: Injection Moulding Cost per Product Diagram

Injection Moulding Material Cost

Material cost refers to the total expense incurred to acquire the raw materials including the plastic and the mold needed to manufacture the toy car mold and to carry out the process of the injection molding. For the mould, it was decided to outsource it. Otherwise it would be needed to buy a machine to produce the mould, add it to the assembly line, calculate the machine cost and also add the labour cost. And since the mould production is a one time process in the beginning of the whole production, it was decided to outsource it. This means that for the material cost only the plastic material that was injected to the mould will be calculated (per product), and the mould calculation will be done separately. The mould cost will be added to the whole injection moulding costs (which is injection machine cost per hour multiplied by the production quantity), this will be shown later on in this section. So the mould cost will be calculated later on.

For the plastic material, Sabic PP 576 P (with the product number CBPP in Cube Production) was used. For carrying the calculations, material cost formula is computed:

$$\text{Material cost} = \text{Plastic weight per product (g)} \cdot \text{Material cost per grams (€/g)}$$

Calculation for the material cost can be seen below:

$$\text{Plastic weight per product} = 19,7 \text{ g}$$

$$\text{Material cost per product: } 1000 \text{ kg} = 1360\text{€ (from the price list)} \Rightarrow 19,7 \text{ g} = 0,0268 \text{ €}$$

$$\text{Material cost} = 1,97 \text{ g} \cdot 0,0268 \text{ €/g} = 0,5278 \text{ €}$$

Injection Moulding Primary Machine Cost

After material cost calculation, primary machine calculations comes next. Primary machine cost means the main time the machine works, and for injection moulding that is the time the machine is actively used for the injection moulding process. For injection molding, the primary cost can be calculated with the sum of filling cost (which is associated with filling the mold cavity with molten plastic during the injection phase of the molding cycle) and cooling cost, since filling the mold with the plastic and then cooling the mold are the two main events happening during injection molding. Before calculating the filling and cooling cost, the "injection machine cost per hour (€/hour)" is needed to be calculated, which can be computed with the formula below:

$$\begin{aligned}
 &\textbf{Injection molding machine cost per hour (€/h)} \\
 &= \textbf{Labour cost per hour (€/h) + Depreciation cost (€/h)} \\
 &= 25,1987 + 0,5459 \\
 &= 25,7445 \text{ €/hours}
 \end{aligned}$$

Labour and depreciation costs were already calculated in the section 2.2.3. Next, calculations for filling and cooling costs will be shown. In the calculations, filling time and cooling time were taken from the Moldex simulation, and then they were converted to hours from seconds. Cooling time is 22 seconds and filling time is 1,157 seconds. However, in the calculations because they are converted to hours; so they will be shown in decimals.

$$\begin{aligned}
 &\textbf{Filling cost = Injection machine cost per hour (€/h) · filling time (h)} \\
 &= 24,7445 \cdot 0,0003 \text{ hours (1,157 seconds)} \\
 &= 0,0083 \text{ €}
 \end{aligned}$$

$$\begin{aligned}
 &\textbf{Cooling cost = Injection machine cost per hour (€/h) · cooling time (h)} \\
 &= 24,7445 \cdot 0,006 \text{ hours (22 seconds)} \\
 &= 0,1573 \text{ €}
 \end{aligned}$$

From all this information, the primary machine cost is calculated by adding these two the filling and cooling costs, which can be seen in the calculation below:

$$\textbf{Primary machine cost = 0,0083 + 0,1573 = 0,1656 €}$$

Injection Moulding Secondary Machine Cost

The following step is to calculate the secondary machine cost. Secondary machine cost refers to the cost of supporting equipment and auxiliary machines required to operate the injection molding system efficiently, when there is not an active injection moulding process is happening. These are not directly part of the injection unit but are necessary for the full production process. Calculation of secondary machine costs can be seen below:

$$\begin{aligned}
 &\textbf{Secondary machine cost (€) =} \\
 &\textbf{Injection moulding machine cost per hour (€/h) · Secondary time (h)} \\
 &= 24,7445 \text{ (€/h) · 0,0014 (h) (5 seconds)} \\
 &= 0,0358 \text{ €}
 \end{aligned}$$

Injection Moulding Overhead Cost

The last component that needs to be calculated for the injection machine cost is the overhead cost. Overhead costs are needed for the production line to continue, but cannot really put into the other categories. It composes of rent, utilities, insurance and maintenance. Before calculating these each specific costs, the general formula used for overhead costs can be seen below:

$$\text{Overhead costs (€)} = \text{Overhead costs per hour (€/h)} \cdot [\text{Secondary time (h)} + \text{Cooling time (h)}]$$

Overhead cost per hour is calculated by the sum of rent, utilities, insurance and maintenance costs. The reason having the sum of secondary time and cooling time is to consider the main time the *injection molding machine works*. The main time machine works of consist of primary and secondary time. For secondary machine time it is directly taken from the secondary machine cost calculation, but for the primary machine time the cooling time is taken. Since cooling takes around 22 seconds and filling only takes around 1 second, filling time doesn't contribute to the primary machine time as much as cooling. Now the calculations of each category of overhead cost per hour will be shown below:

$$\text{Rent Cost (€)} = \frac{\text{Annual rent (€)}}{\text{Working hours per year}} = \frac{5000 \text{ €}}{1832} = 27293 \text{ €}$$

$$\text{Utilities Cost (€)} = \frac{\text{Annual utilities (€)}}{\text{Working hours per year}} = \frac{3000 \text{ €}}{1832} = 1,6376 \text{ €}$$

$$\text{Insurance Cost (€)} = \frac{\text{Purchase price of the machine (€)} \cdot 2\%}{\text{Working hours (per year)}} = \frac{10000 \cdot 2\%}{1832} = 0,1092 \text{ €}$$

$$\text{Maintenance Cost (€)} = \frac{\text{Purchase price of the machine (€)} \cdot 5\%}{\text{Working hours (per year)}} = \frac{10000 \cdot 5\%}{1832} = 0,2729 \text{ €}$$

Annual rent and annual utilities costs were assumed while making these calculations. For rent, the annual rent was divided to working hours to find the rent overhead cost per hour. With the same logic, annual utilities were also divided to working hours per year. In the insurance and maintenance costs; annual insurance was assumed to be 2 percent of the machine price, and annual maintenance was assumed to be 5 percent of the machine price. After calculating the annual insurance cost and the annual maintenance cost, they were again divided by the working hours to find each type of overhead cost per hour. Overhead costs per hour is the sum of these 4 costs, which is 4,7489 (€/h). After obtaining these values, overhead costs can be calculated:

$$\begin{aligned} \text{Overhead costs (€)} &= \text{Overhead costs per hour (€/h)} \cdot [\text{Secondary time (h)} + \text{Cooling time (h)}] \\ &= 4,6056 \text{ (€/h)} \cdot [0,0014 \text{ (h)} \text{ (5 seconds)} + 0,0061 \text{ (h)} \text{ (22 seconds)}] \\ &= 0,0371 \text{ €} \end{aligned}$$

Injection Moulding Machine Cost per Product

Within all the calculations that have been done so far, the injection moulding machine cost per product will be calculated as the sum of material cost, primary machine cost, secondary machine cost and overhead cost; as seen below:

$$0,5278 \text{ €} + 0,1656 \text{ €}, 0,0358 \text{ €} + 0,0371 \text{ €} = 0,7663 \text{ €}$$

Injection Moulding Mould Cost

Now that injection moulding machine cost per product is calculated, the remaining component to calculate is the mould cost for injection machine cost. For outsourcing the mould, a research was done to find places the mould could have been outsourced, and Protolabs was chosen from the alternatives. From the website of Protolabs, it was found out that the cost of one mould is 487€ (Protolabs, Manufacturing Accelerated, n.d.). There will be one mould used for the injection moulding, and there will be another one as backup. So there will be 2 mould in total. This means that the total mould cost is 974€.

In appendix A the calculations that have been done so far can be seen in the excel sheet.

Injection Moulding Machine Cost

Now that the injection moulding machine cost per product and the mould cost is known, the injection moulding machine cost can be found. The injection moulding machine cost per product will be multiplied by the production amount, and then the mould cost will be added. Calculations can be seen below:

$$\begin{aligned} & \text{Injection moulding machine cost} \\ &= (\text{Injection moulding machine cost per product} \cdot \text{production amount}) + \text{mould cost} \\ &= (0,7663 \cdot 200000) + 974 \\ &= 154234,7059 \text{ €} \end{aligned}$$

2.2.5 Vacuum Forming Machine Cost

Vacuum forming machine cost is calculated in the same way as injection moulding, by multiplying vacuum forming machine cost per product with the production quantity, which is 200000. Again the vacuum forming machine cost per product is calculated by the material cost, primary machine cost, secondary machine cost and overhead cost all added to each other. An overview of the vacuum forming machine cost per hour can be seen in the figure below:

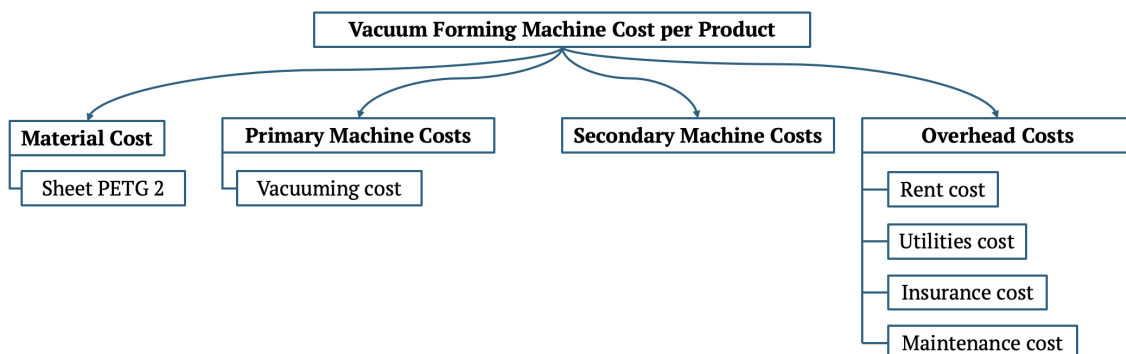


Figure 21: Vacuum Forming Cost per Product Diagram

Vacuum Forming Material Cost

Material cost for vacuum forming, as similar as injection moulding it includes the mould cost and the plastic material cost. For the mould, it was decided to outsource it because of the same reason that was addressed in injection moulding. This means that for the material cost only the plastic sheets that will be shaped on top of the mould will be calculated (per product), and the mould calculation will be done separately, then added to the whole vacuum forming costs (which is vacuum forming machine cost per hour multiplied by the production quantity).

For the plastic sheets, sheet PETG2 (with the product number CBPG2 in Cube Production) was used. In the product list, one sheet is said to be 6,90 €, and the dimensions are 1000 mm x 1000 mm. For one car body, the sheet dimension needed is 30 mm x 30 mm. Which means that from one sheet bought, 900 car bodies can be produced. Thus to find the sheet cost of one car 6,99 € is divided to 900, and that makes **0,2091 €**.

Vacuum Forming Primary Machine Cost

The next category of vacuum forming machine cost to calculate is the primary machine cost. For vacuum forming, the primary machine cost can be calculated by the vacuuming cost, since it is the main event in this process. Before calculating the vacuuming cost, the "vacuum forming cost per hour (€/hour)" is needed to be calculated, which can be computed with the formula below:

$$\begin{aligned}
 &\textbf{Vacuum forming machine cost per hour (€/h)} \\
 &= \textbf{Labour cost per hour (€/h) + Depreciation cost (€/h)} \\
 &= 50,3974 + 1,0917 \\
 &= 51,4891 \text{ €/hours}
 \end{aligned}$$

Labour and depreciation costs were already calculated in the section 2.2.3. Next, the calculation for vacuuming cost can be seen below, which makes up the primary machine cost. For vacuuming time, it was stated that it is 60 seconds in the production part of the report, which can be found in section 1.3. 60 seconds were converted to hours during the calculations, so they will be shown with decimals:

$$\begin{aligned}
 &= \textbf{Primary machine cost} \\
 &= \textbf{Vacuuming cost} \\
 &= \textbf{Vacuum forming machine cost per hour (€/h) \cdot Vacuuming time (h)} \\
 &= 51,4891 \cdot 0,008 \text{ hours (60 seconds)} \\
 &= \textbf{0,0429 €}
 \end{aligned}$$

Vacuum Forming Secondary Machine Cost

Moving on to the secondary machine cost, the same formula is used as the injection moulding, and the secondary time is the same as well. Calculation for the secondary machine cost can be seen below:

$$\begin{aligned}
 \textbf{Secondary machine cost (€)} &= \textbf{Vacuum forming machine cost per hour (€/h) \cdot Secondary time (h)} \\
 &= 51,4891 \text{ (€/h)} \cdot 0,0014 \text{ hours (5 seconds)}
 \end{aligned}$$

$$= 0,0715 \text{ €}$$

Vacuum Forming Overhead Cost

Lastly, the overhead costs are left for vacuum forming. The way overhead costs are calculated is the same, however, there is a difference due to the different purchase price of the machines. Calculations of the overhead costs can be seen below:

$$\text{Rent Cost (€)} = \frac{\text{Annual rent (€)}}{\text{Working hours per year}} = \frac{5000 \text{ €}}{1832} = 2,7293 \text{ €}$$

$$\text{Utilities Cost (€)} = \frac{\text{Annual utilities (€)}}{\text{Working hours per year}} = \frac{3000 \text{ €}}{1832} = 1,6376 \text{ €}$$

$$\text{Insurance Cost (€)} = \frac{\text{Purchase price of the machine (€)} \cdot 2\%}{\text{Working hours (per year)}} = \frac{20000 \cdot 2\%}{1832} = 0,2183 \text{ €}$$

$$\text{Maintenance Cost (€)} = \frac{\text{Purchase price of the machine (€)} \cdot 5\%}{\text{Working hours (per year)}} = \frac{20000 \cdot 5\%}{1832} = 0,5459 \text{ €}$$

Overhead costs are the sum of these four categories, which is 5,1310 € for vacuum forming. Overhead costs per hour can be calculated as shown below:

$$\begin{aligned} \text{Overhead costs (€)} &= \text{Overhead costs per hour (€/h)} \cdot [\text{Secondary time (h)} + \text{Vacuuming time (h)}] \\ &= 5,1310 \text{ (€/h)} \cdot [0,0014 \text{ hours (5 seconds)} + 0,0008 \text{ hours (60 seconds)}] \\ &= 0,0114 \text{ €} \end{aligned}$$

Vacuum Forming Machine Cost per Product

Within all the calculations that have been done so far, the vacuum forming machine cost per product will be calculated as the sum of material cost, primary machine cost, secondary machine cost and overhead cost; as seen below:

$$0,2091 \text{ €} + 0,0429 \text{ €, } 0,0715 \text{ €} + 0,0114 \text{ €} = 0,3349 \text{ €}$$

Vacuum Forming Mould Cost

Now that vacuum forming cost per product is calculated, the remaining component to calculate is the mould cost for injection machine cost. For outsourcing the mould, same research as injection moulding was done to find places the mould could have been outsourced, and Rapid Prototyping NL was chosen from the alternatives. The reason for choosing a different company than the injection moulding is that moulds for two different machines are produced with different production types. For injection moulding mould milling is used, but for the vacuum forming mould 3D printing is used. And different companies had different selections of production. From this websire, it was found out that the cost of one mould is 67€ (RAPIDPrototyping.nl, n.d.). One mould will be used for 200 products, as the mould can to be deformed from the heat. Considering that 200000 cars are produced, 1000 moulds

will be needed. This means that the total mould cost is 67000€.

In appendix B these calculations can be seen in the excel sheet.

Vacuum Forming Machine Cost

Now that the vacuum forming cost per product and the mould cost is known, the vacuum forming cost can be found. Same logic as injection moulding will be applied, vacuum forming cost per product will be multiplied by the production amount, and then the mould cost will be added. Calculations can be seen below:

$$\begin{aligned}
 &\textbf{Vacuum forming cost} \\
 &= (\textbf{Vacuum forming cost per product} \cdot \textbf{production amount}) + \textbf{mould cost} \\
 &= (0,3349 \cdot 200000) + 67000 \\
 &= \mathbf{133982,6651 \text{ €}}
 \end{aligned}$$

2.2.6 Laser Cutting Machine Cost

Laser cutting machine cost is again calculated in the method; by multiplying laser cutting machine cost per product with the production quantity, which is 200000. Again the laser cutting machine cost per product is calculated by the material cost, primary machine cost, secondary machine cost and overhead cost all added to each other. An overview of the laser cutting machine cost per hour can be seen in the figure below:

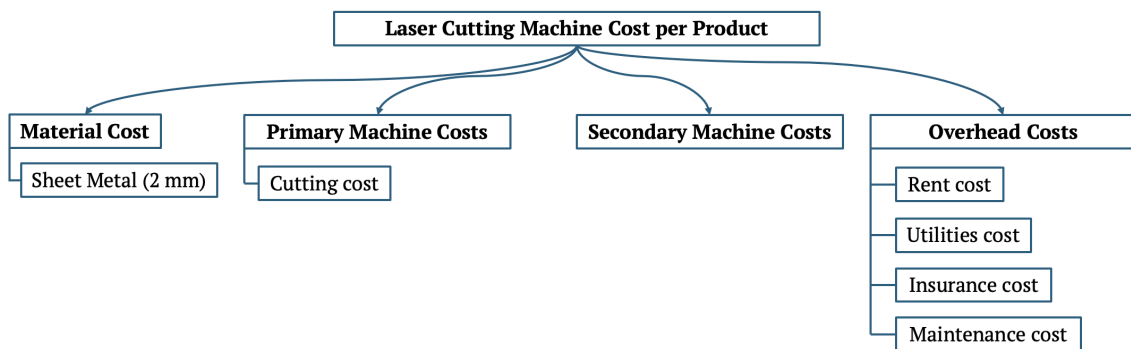


Figure 22: Laser Cutting Cost per Product Diagram

Laser Cutting Material Cost

Material cost for laser cutting refers to costs of sheets. The material is sheet metal with 2mm thickness (with the production code CBSM2 in Cube production), with the dimensions 2000 mm x 1000 mm. Cost of one sheet metal is 94,2 €. Considering the dimensions of the bottom plate, there can be 272 pieces produced per sheet, and since 200000 of them are needed there will be 735 sheets needed. And to calculate the material cost per product, the total sheet cost will be divided by the production amount, which is 200000. Equation of this calculation can be seen below:

$$\textbf{Material cost per product: } \frac{94,2 \text{ €} \cdot 735 \text{ sheets}}{200000 \text{ pieces}} = \mathbf{0,3462 \text{ €}}$$

Laser Cutting Primary Machine Cost

Next section to make calculations is primary machine cost. For laser cutting, the primary cost can be calculated by the cutting cost. Before calculating it, "laser cutting cost per hour (€/hour)" is needed to be calculated, which can be computed with the formula below:

$$\begin{aligned}
 & \textbf{Laser cutting machine cost per hour (€/h)} \\
 &= \textbf{Labour cost per hour (€/h) + Depreciation cost (€/h)} \\
 &= 25,1987 + 7,5328 \\
 &= 32,7314 \text{ €/hours}
 \end{aligned}$$

Labour and depreciation costs were already calculated in the section 2.2.3. Next, the calculation for cutting cost is needed for using it in the primary machine cost. For the cutting cost calculation, cutting time is needed. The cutting time was founded from the cut length and the cutting speed. The cut length is 390 mm, which was obtained from the bottom plate design. Since the laser cutting machine cost per product is calculated, only 1 bottom plate's dimensions were considered for the cut length. The cutting speed was estimated to be 10 m/min. From these values and also making the proper unit conversions, the cut time was found to be *0,007 hours*. After finding the cutting time, the cutting cost and thus primary machine cost can be calculated:

$$\begin{aligned}
 &= \textbf{Primary machine cost} \\
 &= \textbf{Cutting cost} \\
 &= \textbf{Laser cutting machine cost per hour (€/h) \cdot Cutting time (h)} \\
 &= 32,7314 \cdot 0,007 \\
 &= \textbf{0,0213 €}
 \end{aligned}$$

Laser Cutting Secondary Machine Cost

Next calculation will be the secondary machine cost. The same formula is used as as the injection moulding and vacuum forming, also same secondary time was used as well. Calculation for the secondary machine cost can be seen below:

$$\begin{aligned}
 \textbf{Secondary machine cost (€)} &= \textbf{Laser cutting machine cost per hour (€/h) \cdot Secondary time (h)} \\
 &= 32,7314 \text{ (€/h)} \cdot 0,0014 \text{ (h)} \\
 &= \textbf{0,0455 €}
 \end{aligned}$$

Laser Cutting Overhead Cost

Last calculations will be for the overhead costs. The way overhead costs are calculated again the same with a difference of different purchase price of the machine. Calculations of the overhead costs can be seen below:

$$\textbf{Rent Cost (€)} = \frac{\textbf{Annual rent (€)}}{\textbf{Working hours per year}} = \frac{5000 \text{ €}}{1832} = 2,7293 \text{ €}$$

$$\text{Utilities Cost (€)} = \frac{\text{Annual utilities (€)}}{\text{Working hours per year}} = \frac{3000 \text{ €}}{1832} = 1,6376 \text{ €}$$

$$\text{Insurance Cost (€)} = \frac{\text{Purchase price of the machine (€)} \cdot 2\%}{\text{Working hours (per year)}} = \frac{138000 \cdot 2\%}{1832} = 1,5066 \text{ €}$$

$$\text{Maintenance Cost (€)} = \frac{\text{Purchase price of the machine (€)} \cdot 5\%}{\text{Working hours (per year)}} = \frac{138000 \cdot 5\%}{1832} = 3,7664 \text{ €}$$

Overhead costs are the sum of these four categories, which is 9,6397 € for vacuum forming. Overhead costs per hour can be calculated as shown below:

$$\begin{aligned} \text{Overhead costs (€)} &= \text{Overhead costs per hour (€/h)} \cdot [\text{Secondary time (h)} + \text{Cutting time (h)}] \\ &= 9,6397 \text{ (€/h)} \cdot [0,0014 \text{ (h)} + 0,0007 \text{ (h)}] \\ &= 0,0197 \text{ €} \end{aligned}$$

Laser Cutting Machine Cost per Product

Within all the calculations that have been done so far, the laser cutting machine cost per product will be calculated as the sum of material cost, primary machine cost, secondary machine cost and overhead cost; as seen below:

$$0,3462 \text{ €} + 0,0213 \text{ €, } 0,0455 \text{ €} + 0,0197 \text{ €} = 0,4326 \text{ €}$$

In appendix C the calculations that have been done so far can be seen in the excel sheet.

Laser Cutting Machine Cost

Now that the laser cutting machine cost per product is known, the laser cutting machine cost can be found. Same approach as vacuum forming and injection moulding will be done again, but without the mould cost since there is no mould in laser cutting. Laser cutting machine cost per product will be multiplied by the production amount. Calculations can be seen below:

$$\begin{aligned} &\text{Laser cutting machine cost} \\ &= (\text{Laser cutting machine cost per product} \cdot \text{production amount}) \\ &= (0,4326 \cdot 200000) \\ &= 86515,0252 \text{ €} \end{aligned}$$

2.2.7 CNC Router Machine Cost

In the section 1.6.4, some new steps were mentioned; and cutting the sheets for vacuum forming as well as trimming the edges of the sheets after vacuum forming were two of the new steps. In order for these steps to be done, a machine is needed. Therefore a CNC router machine is also decided to be bought. The machine cost is calculated within the same method as the previous machines, by multiplying machine cost per product with the production quantity, which is 200000. However, the machine cost product calculation is a little more different. Since this machine will only cut the plastic sheets for vacuum forming, there will not be a material cost. The material cost of the plastic vacuum forming sheets were already calculated. But apart from that; primary machine cost, secondary machine cost and overhead costs have the same approach. An overview of the CNC router machine cost per hour can be seen in the figure below:

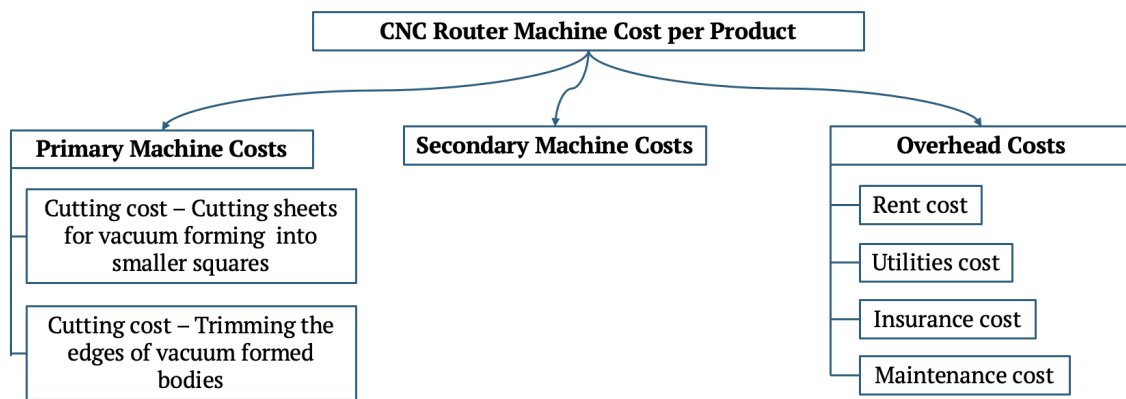


Figure 23: CNC Router Cost per Product Diagram

CNC Router Primary Machine Cost

Next section to make calculations is primary machine cost. For CNC router, the primary cost will be calculated by the cutting cost of two processes: Cutting sheets for vacuum forming into smaller squares and trimming the edges of vacuum formed bodies. Before calculating it, "CNC router cost per hour (€/hour)" is needed to be calculated, which can be computed with the formula below:

$$\begin{aligned}
 &\text{CNC router machine cost per hour (€/h)} \\
 &= \text{Labour cost per hour (€/h)} + \text{Depreciation cost (€/h)} \\
 &= 25,1987 + 2,7293 \\
 &= 27,9279 \text{ €/hours}
 \end{aligned}$$

Labour and depreciation costs were already calculated in the section 2.2.3. Next, the calculation for two cutting costs are needed for using it in the primary machine cost. For the cutting cost calculation, cutting time is needed. The cutting times were estimated in the 1.6.4 section. For cutting sheets for vacuum forming into smaller squares, 10 seconds are needed. For trimming the edges of vacuum formed bodies, 15 seconds are needed. In the calculations hours are needed. Within the necessary conversion, the cutting time is 0,0028 hours and trimming time is 0,0042 hours. After finding the cutting and trimming times, the total cutting cost and thus primary machine cost can be calculated:

$$\begin{aligned}
&= \text{Primary machine cost} \\
&= \text{Total Cutting cost} \\
&= \text{CNC router cutting machine cost per hour (€/h)} \cdot \text{cutting time (h)} + \\
&\quad \text{CNC router cutting machine cost per hour (€/h)} \cdot \text{trimming time (h)} \\
&= 27,9279 \cdot 0,0028 + 27,9279 \cdot 0,0042 \\
&= \mathbf{0,1939 \text{ €}}
\end{aligned}$$

CNC Router Secondary Machine Cost

Next calculation will be the secondary machine cost. The same formula is used as as the injection moulding and vacuum forming, also same secondary time was used as well. Calculation for the secondary machine cost can be seen below:

$$\begin{aligned}
\text{Secondary machine cost (€)} &= \text{CNC router machine cost per hour (€/h)} \cdot \text{Secondary time (h)} \\
&= 27,9279 \text{ (€/h)} \cdot 0,0014 \text{ (h)} \\
&= \mathbf{0,0388 \text{ €}}
\end{aligned}$$

CNC Router Overhead Cost

Last calculations will be for the overhead costs. The way overhead costs are calculated again the same with a difference of different purchase price of the machine. Calculations of the overhead costs can be seen below:

$$\text{Rent Cost (€)} = \frac{\text{Annual rent (€)}}{\text{Working hours per year}} = \frac{5000 \text{ €}}{1832} = 2,7293 \text{ €}$$

$$\text{Utilities Cost (€)} = \frac{\text{Annual utilities (€)}}{\text{Working hours per year}} = \frac{3000 \text{ €}}{1832} = 1,6376 \text{ €}$$

$$\text{Insurance Cost (€)} = \frac{\text{Purchase price of the machine (€)} \cdot 2\%}{\text{Working hours (per year)}} = \frac{50000 \cdot 2\%}{1832} = 0,5459 \text{ €}$$

$$\text{Maintenance Cost (€)} = \frac{\text{Purchase price of the machine (€)} \cdot 5\%}{\text{Working hours (per year)}} = \frac{50000 \cdot 5\%}{1832} = 1,3646 \text{ €}$$

Overhead costs are the sum of these four categories, which is 6,27 € for vacuum forming. Overhead costs per hour can be calculated as shown below:

$$\begin{aligned}
\text{Overhead costs (€)} &= \text{Overhead costs per hour (€/h)} \cdot [\text{Secondary time (h)} + \text{Total cutting time (h)}] \\
&= 6,2773 \text{ (€/h)} \cdot [0,0014 \text{ (h)} + 0,0042 \text{ (h)} + 0,0028 \text{ (h)}]
\end{aligned}$$

$$= 0,0523 \text{ €}$$

CNC Router Machine Cost per Product

Within all the calculations that have been done so far, the CNC router machine cost per product will be calculated as the sum of primary machine cost, secondary machine cost and overhead cost; as seen below:

$$0,1939 \text{ €} + 0,0388 \text{ €} + 0,0523 \text{ €} = 0,2850 \text{ €}$$

In appendix [D](#) the calculations that have been done so far can be seen in the excel sheet.

CNC Router Machine Cost

Now that the CNC router machine cost per product is known, the CNC router machine cost can be found. Same approach as as before will be done again, CNC router machine cost per product will be multiplied by the production amount. Calculations can be seen below:

CNC router machine cost

$$\begin{aligned} &= (\text{CNC router machine cost per product} \cdot \text{production amount}) \\ &= (0,2850 \cdot 200000) \\ &= 57008,7336 \text{ €} \end{aligned}$$

2.3 Assembly and packaging

Assembly and packaging costs represent the final category in the overall cost analysis. To determine the cost per toy car, both the manual assembly time and the cost of packaging materials were estimated using data from the workstations and packaging suppliers. The following section will first address the assembly costs, followed by the packaging costs.

2.3.1 Assembly cost

For the assembly costs, the main approach is to calculate the *total working days needed to finish all the assembling process* and the *employee cost per day*. After finding these two values, they will be multiplied and the total assembly cost will be found. From that point, the assembling cost per car will be found. This process will be explained in detail in this section of the report.

2.3.2 Total working days needed to finish all the assembling process

For the assembly costs, firstly total working days needed to finish all the assembling process will be found. To find it, the assembly time is needed; and thus the workstation times are needed.

Assembly Line Components

The assembly process is divided into four workstations, each operated by one employee. And there are additional workstations that were added in the section [1.6.4](#) as well, which are shown as the "additional stations" in the table:

Stations	Task Description	Estimated Time
WS1	Cutting the sharp edges of the wheels	~22 seconds
WS2	Fixing the wheels to the shaft	~26 seconds
WS3	Attaching shafts and partly assembling the frame	~23 seconds
WS4	Final assembly and snapping the car body	~25 seconds
Additional Station 1	Cutting 2 shafts	~10 seconds
Additional Station 2	Trimming edges of vacuum formed sheet	~15 seconds

Table 10: Workstation breakdown and estimated assembly time

From this table the total assembly time can be found by having a sum of each station's estimated time:

$$22 + 26 + 23 + 25 + 10 + 15 = 121 \text{ seconds, } \frac{121}{3600} = 0.0336 \text{ hours per car}$$

Man Hour Calculations

After showing the assembly line, man hour calculations will be made. Man hour is used in the assembly and packaging costs (used generally in production). It is a unit of measurement that represents the amount of work one person can complete in one hour. For a full-time job of an employee, the common man hours in the Netherlands (country where this toy car is being manufactured) are 8 hours. In other words, one person works for 8 hours a day. Calculating the man hours is a key part to calculate the total working days needed to finish the assembling process. It was calculated that the total amount of hours needed to produce the 200,000 cars in a year was **6722,22 hours**:

$$\begin{aligned} \text{Total time needed to assemble cars} &= \frac{\text{Total time per car (seconds)} \cdot \text{Production amount}}{3600} \\ &= \frac{121 \text{ seconds} \cdot 200000}{3600} = \mathbf{6722,22 \text{ hours}} \end{aligned}$$

In the calculation, total time needed to assemble cars is multiplied by the total number of cars that needed to be produced to find the total production time. It was divided by 3600 to get the time in hours (seconds were used before for the workstations). However, in real-life applications, the number of hours will be more than the value found in this calculation (6722,22 hours) due to operational factors and breaks. This situation will be analyzed in the following section.

Operational Factors

There are some delays that might happen during the assembly process. As observed when producing the sample in the workshop, sometimes human errors can occur and make the assembly time longer than expected. Even though steps were well planned, there indeed is a possibility of unexpected events to be happening. These delays fall under operational factors. They influence not only productivity but also cost, as the time spent increases.

An additional 10% of time was included to account for operational factors. These factors, such as delays and inefficiencies, can fluctuate over time. Initially, delays are more frequent, but with repetition and experience, their impact tends to decrease. Nevertheless, they cannot be entirely eliminated due to human error. To reflect this variability, a 10% increase was adopted as an average adjustment. With this operational buffer considered, the total hours required increase to:

$$6272,22 \times 1,10 = 672,22 \text{ hours more}$$

$$\text{Total production time} = 6722,22 + 672,22 = \mathbf{7394,44 \text{ hours}}$$

Breaks

It was mentioned earlier that each working day would be 8 hours. Among the working hours, employees must have minor breaks and lunch breaks, making up to about 1 hour. Since no work is done during this hour, this creates an additional delay. Breaks will add 12,5% to the hours needed:

$$\text{Break Factor} = \frac{1}{8} = 12,5\%$$

Adding breaks to the operational factor-adjusted hours:

$$7394,44 + (7394,44 \times 0,125) = 8318,75 \text{ hours}$$

This value is the total manned time, which means the total time it will take for the employees to work. The total production time does not consider the breaks, thus the total manned time was calculated to have a more accurate result.

Now that the manned hours are known, total man days need to be found. This value was found by dividing the total man hours to 8 hours, which is the time of 1 working day. And this calculation gives **1040 days** for the total manned days. If only 1 employee is working, it would have took around 1040 days to finish the whole work. However, this is not the actual amount of production days to finish the assembling process because there are 5 employees working in total, the 1040 hours are divided among 5 employees. But during the assembly cost calculations, since the total cost of the assembly is wanted to be found the total man days will not be divided among the employees. The calculations will just continue using the *total* manned days. Thus, the total assembly line cost and the assembly cost per unit can be found using the total man days:

$$\text{Total assembly line cost} = \text{Employee cost per day} \cdot \text{total man days}$$

In order to compute this formula, the employee cost per day is needed. For finding employee cost per day, employee cost per hour was multiplied by 8, since there are 8 working hours a day. And the employee cost used in the machine costs were used, which is 25,1987 €. So, the employee cost per day is **201,5895 €**. Now that the employee cost per day is known, the total assembly line cost can continue:

$$\text{Total assembly line cost} = \text{Employee cost per day} \cdot \text{total man days}$$

$$= 201,5895 \cdot 1040 = \mathbf{209621,6021 \text{ €}}$$

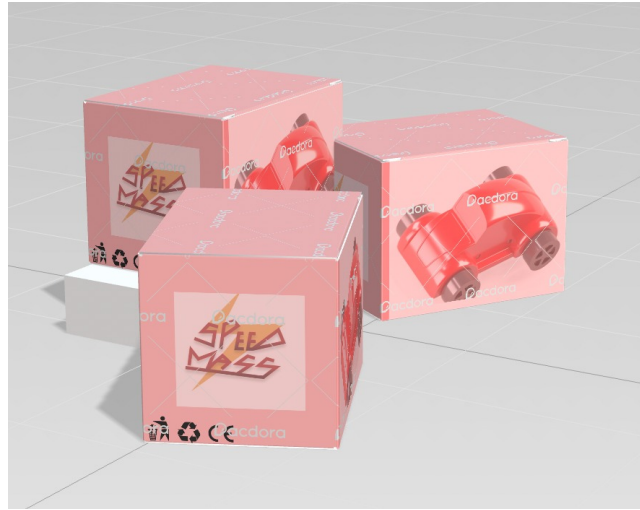
And thus, assembly cost per car is:

$$\text{Assembly cost per car} = \frac{\text{Total assembly line cost (€)}}{\text{Production amount}} = \frac{209621,6021 \text{ €}}{200000} = \mathbf{1,0481 \text{ €}}$$

In appendix **F** these calculations can be seen in the excel sheet.

2.3.3 Packaging cost

After the assembly cost, the next part to calculate is the packaging cost. For the packaging, there were 2 options: to outsource or insource it. Both situations have their advantages and disadvantages, thus both scenarios were calculated to see which one was cheaper. In this section, first, the insourcing scenario will be explained (see subsection 2.3.4) in detail and then, the outsourcing (see subsection 2.3.5). The packages are expected to look similar to the image below at the end of the packaging process, regardless of the scenarios chosen:



2.3.4 Insourced packaging scenario

Insourcing the packaging requires two categories of calculation to be considered: *labour cost of the packaging process* and *the material cost of the packaging*. First the labour cost of the packaging process will be calculated. In this case, packaging will be performed by two employees, at two workstations. The first packaging employee folds the net cardboard into a box. Second employee places the assembled toy car into the box, closes it and glues it. The reason why 2 workstations were used in the insourcing condition is to enhance the efficiency in terms of time. It was also considered to make one employee responsible for both assembling and glueing the packaging as well as putting the cars in it. However, even though two employees would seem costly upfront, the increased output per hour offsets. This means that more packaging will be produced in less time, which reduced the labor cost per unit. The material of the packaging itself will be bought externally, in which the cardboard boxes will be delivered pre-cut to the required dimensions.

Packaging Labour Cost

The first employee will be working in packaging station 1, and the second employee will be working in packaging station 2. Packaging station 1 is estimated to be taking *20 seconds*, and packaging station 2 is estimated to be taking *10 seconds*. Thus, total packaging time can be found as 30 seconds, and in terms of hours it is 0,0083 hours. From this information, the packaging labour cost per day can be calculated as:

$$\begin{aligned} \text{Packaging labour cost per day} &= \text{Total packaging time (hours)} \cdot \text{Employee cost per hour} \\ &= 0,0083 \cdot 25,1987 = \mathbf{0,21 \text{ €}} \end{aligned}$$

Packaging Material Cost

After the packaging labour cost, the next category to calculate is the packaging material cost, which

requires calculating the amount of material needed for one package. To do that, the dimensions of the car is needed, which was calculated from the car:

Car Dimensions

- **Car length (mm)**= 126 mm
- **Car width (mm)**= 72 mm
- **Car height (mm)**= 61 mm

To find the packaging dimensions, twenty percent was added to the car's dimensions. There are several reasons for adding a 20% area margin. The first reason is due to *gluing and folding flaps*. When designing the packaging box, the unfolded package requires additional flaps beyond the six sides of the box. These flaps are necessary to glue the sides together during assembly, reinforce the joints, and ensure that the box holds its shape reliably. Without these flaps, it would not be possible to assemble the packaging box, as cardboard material is generally less rigid than plastic or other materials. The next reason is the *tolerances for cutting and alignment*. Minor errors can occur during cutting due to slight expansion or contraction of the material. Additionally, folding lines may require extra space to prevent cracking or bulging. And the last reason is the *safety allowance*. A safety margin accommodates small deformations of the material, ensuring that all parts of the net align correctly when folded. So, within the added twenty percent the package size can be seen below:

Package Dimensions

- **Length (L)** = 151 mm
- **Width (W)** = 86 mm
- **Height (H)** = 73 mm

From the package dimensions, the package volume can be calculated:

$$V = 151 \text{ mm} \times 86 \text{ mm} \times 73 \text{ mm}$$

$$V = 956261 \text{ mm}^3$$

The total internal volume of the package is:

$$V = 956,261 \text{ mm}^3$$

This value represents the internal space available inside one packaging box for the toy car.

Now that the package volume is known, the packaging material cost per car will be calculated by calculating the number of packages that can be produced from one packaging sheet. The packaging material was chosen to be chipboard, which is one of the most used materials. For doing this calculation, the surface area of one package is needed firstly. The surface area of a the package is calculated with the formula below:

$$\begin{aligned} \text{Surface Area} &= 2(LW + LH + WH) \\ &= 2(151 \cdot 86 + 151 \cdot 73 + 86 \cdot 73) = 60912 \text{ mm}^2 \end{aligned}$$

Now that the surface area of a package is known, the area of one chipboard is needed. From there, the amount of boxes that can fit into the chipboard sheet can be found. A standard chipboard sheet has dimensions:

$$2440 \text{ mm} \times 1220 \text{ mm}$$

$$\text{Area of One Sheet} = 2440 \times 1220 = 2976800 \text{ mm}^2$$

Now, the estimated number of boxes per Sheet (without layout optimization) can be found, as shown below:

$$\text{Number of boxes needed} = \frac{\text{Sheet Area}}{\text{Package Surface Area}} = \frac{29768000}{60912} = 48,8705$$

So, approximately 48 packages can be produced from one chipboard sheet. From there, total sheets needed for 200,000 Packages can be found:

$$\text{Total sheets needed} = \frac{\text{Production amount}}{\text{Boxes that can produced from one sheet}} = \frac{200,000}{48} = 4166,6667$$

This means approximately 4167 sheets are needed to make a package for every single box. Within the amount of sheets needed, their costs can be calculated, as shown below:

$$\text{Price per sheet: €26.94, Total sheets needed: 4167}$$

$$\Rightarrow \text{Total Material Cost} = \text{Price per sheet} \cdot \text{Amount of sheets needed}$$

$$4167 \cdot 26.94 = 112258,9800 \text{ €}$$

Thus, the packaging material cost per car can be found by dividing the total packaging material cost to the production amount:

$$\begin{aligned} \text{Packaging material cost per car} &= \frac{\text{Total packaging material cost}}{\text{Production amount}} \\ &= \frac{112258,9800}{200,000} = 0,5613 \text{ € per car (insourced scenario)} \end{aligned}$$

2.3.5 Outsourcing packaging scenario

After finding the packaging cost per car for insourcing scenario, the outsourced scenario will also be calculated. To evaluate the cost of outsourcing packaging, the parcel shipping costs for transporting 200000 packaging boxes will be calculated. The shipping cost includes both shipping the toy cars to the place that they will be packaged, and shipping the packages back to the factory. First of all, the total space that the packages will keep is needed. To find the total space, one package size is needed. In the insourcing scenario, it was found that one package is *956261 milimeter cube*. However, the

units will be meter for this calculation, and converting 956261 to meters gives *0,0010 meter cube*. Considering that there will be 200000 packages, the total space needed is:

$$\text{Total Space needed} = 200000 \times 0,0010 = 191,2523 \text{ m}^3$$

For the shipping, the Quicargo website was used to take shipping values as reference. From the website, 1 parcel of 1 meters x 0,5 meters x 0,5 meters costs 9,17 €. To this parcel, 261 boxes can fit, which can be seen in the calculations below:

$$\text{Boxes per Parcel} = \frac{\text{Parcel size}}{\text{Package size}} = \frac{1 \text{ m} \cdot 0,5 \text{ m} \cdot 0,5 \text{ m}}{0,001 \text{ m}} = \mathbf{261,4348 \text{ boxes}}$$

Thus, total number of parcels needed required can be found by dividing the production amount to the amount of boxes that can fit into one parcel:

$$\text{Amount of parcels needed} = \frac{\text{Production amount}}{\text{Boxes that can fit into one parcel}} = \frac{200000}{261} = \mathbf{766 \text{ parcels needed}}$$

Finally, the total shipping cost and shipping cost per car can be calculated:

$$\begin{aligned} \text{Total shipping cost} &= \text{Packaging shipping cost per parcel} \cdot \text{Amount of parcels needed} \\ &= 766 \cdot 9,17 = \mathbf{7026,82 \text{ €}} \end{aligned}$$

$$\text{Shipping cost per car} = \frac{\text{Total shipping cost}}{\text{Production amount}} = \frac{7026,82}{200,000} = \mathbf{0,0351 \text{ €}}$$

In appendix [G](#) these calculations can be seen in the excel sheet.

2.3.6 Evaluation of packaging cost scenarios

After calculating each scenario, the insourced and outsourced packaging can be compared. Total cost per toy car when packaging is **insourced** is:

$$\text{Assembly Cost} + \text{Insourced Packaging Cost} = \mathbf{1,8194 \text{ €}}$$

$$\text{Assembly Cost} + \text{Outsourced Packaging Cost} = \mathbf{1,0832}$$

If the outsourcing scenario is considered, a reduction in costs is observed. Therefore, SpeedMass decided to go with the outsourcing scenario to save on employee costs and benefit from industry-standard packaging practices.

In appendix [H](#) these calculations can be seen in the excel sheet.

2.4 Mass production cost and cost per toy car

In this section everything that has been calculated will be summed up and a final cost for the mass production and cost per car will be shown. First, the total set up cost was calculated:

$$\begin{aligned}
 \text{Total set up costs} &= \text{Injection moulding set up} + \text{Vacuum forming set up} \\
 &+ \text{Laser cutting set up} + \text{Software (CAM, Moldex, Flexsim) costs} + \text{Designer Cost} \\
 &= 75,7445 \text{ €} + 1151,4891 \text{ €} + 132,7314 \text{ €} + 1500 \text{ €} + 480 \text{ €} \\
 &= \mathbf{2339,9651 \text{ €}}
 \end{aligned}$$

Next and final step is to add the rest of the costs calculated, which can be computed with the formula below:

$$\begin{aligned}
 \text{Total cost (mass production)} &= \text{Set up costs} + \text{Injection moulding machine cost} \\
 &+ \text{Vacuum forming machine cost} + \text{Laser cutting machine cost} + \text{CNC router machine cost} \\
 &+ \text{Ordered parts material cost} + \text{Packaging and assembly cost} \\
 &= 2359,9651 \text{ €} + 154234,7059 \text{ €} + 133982,6651 \text{ €} \\
 &+ 86515,0252 \text{ €} + 558395,2329 \text{ €} + 57008,7336 \text{ €} + 216648,4220 \text{ €} \\
 &= \mathbf{1209144,7498 \text{ €}}
 \end{aligned}$$

From this, cost for one single toy car can be calculated by dividing to 200000:

$$\frac{1209144,7498 \text{ €}}{200000} = \mathbf{6,0457 \text{ €}}$$

In appendix I these calculations can be seen in the excel sheet.

2.5 Evaluation

This section of the report focuses on evaluating the cost estimation strategy used in the production of the SpeedMass toy car. Several types of cost estimations were applied such as machine costs, material costs, labour cost, packaging cost, assembly cost, and much more. Together, these estimations provided a rough approximation of the final product cost.

2.5.1 General evaluation on cost estimation

The cost analysis was conducted as part of the recent progress in the development of SpeedMass. Understanding the company's current position relative to competitors in the market is essential. This insight will be critical in shaping future strategies to enhance the product and reduce the risk of financial loss. Moreover, by knowing the limitations of the strategy, Profit margin and markup (percentage difference between the cost of producing a good and its selling price) can be calculated.

To begin with, key assumptions made will be evaluated. The cost estimation underwent several assumptions, it cannot be said that there was one major category that was assumed, but little by little

all the categories added up for the estimations made. For instance lifespans for all of the machines were estimated, while consulting the employees in the cube it was observed that the minimum life span of the machines was 10 years. Data used for the machine purchase were computed according to the year they were approximately purchased (while consulting with the experts in the Cube, an approximation were given for all the machines' lifespans). This keeps the costs of the machines up to date as much as possible and accurate to the actual cost. Other parts about the machines were estimated like the tooling costs, secondary time (like cleaning, clamping, etc), and overhead costs. However, all the costs that were assumed, affected a small part of the total manufacturing cost. One of the most important parts about the machines that was estimated can be considered as the time considered in the primary machine costs. For vacuum forming, this would be the cooling time and for the laser cutting, this would be the cutting time. It was challenging to estimate this cost information due to the leakage of the softwares used in estimating these formulas. Another important part for the cost analysis can be considered as the purchase price of the machines. In the beginning, there was market research done for the purchase of the machines. This research ended up not being accurate after consulting the actual machines that were used in Cube, and the cost of one single car is did change a lot after entering the actual purchase price of the machines.

Overall, there was generally a logical numerical consistency in the values obtained. End costs of the machines and the materials make sense within the context of the market, as they the machines were searched from their specific models and the materials' prices were obtained from Cube production price list. Something that was observed that vacuum forming shows a relatively lower cost compared to the other manufacturing processes. This difference is primarily attributed to the assumed costs made during the calculations. However, this assumption introduces a degree of uncertainty for the company into the overall cost estimations and may affect the accuracy of the rest of the analysis. Considering that laser cutting also had more assumptions, the same situation is also applicable for laser cutting. For injection moulding, due to having a simulation there were less assumptions made. A simulation may also not considered as one hundred percent true as well, since it it not real life, but it does provide a good starting point and more accurate assumptions.

2.5.2 Breakeven analysis

After making a general analysis on the cost estimation, next analysis that will be made is the breakeven analysis. Breakeven analysis is a financial tool to determine the point at which the toy car covers its costs and begins to generate profit. Bearing in mind that all the products are sold would help in estimating whether the final toy car price brings profit or loss. This is a crucial part of the evaluation process because the breakeven point would be the level of output at which total revenue equals total costs (both fixed and variable). At this point, there is no profit or loss. It is crucial to have such input because this would put a selling goal for the SpeedMass.

The break-even point (BEP) in units is calculated using the formula:

$$\text{BEP} = \frac{\text{Fixed Costs}}{\text{Selling Price per Unit} - \text{Variable Cost per Unit}}$$

Given:

- Fixed Costs = 1209144,7498 €
- Selling Price per Unit = 6,04 €

2.5.3 Breakeven point

- Variable Cost per Unit = 0,866 €

$$\text{BEP}_{\text{outsource}} = \frac{120914470}{6,04 - 0,866} = \frac{1209144,7498}{5,174} \approx \boxed{233720 \text{ units}}$$

Break-even suggests that when 233720 units should be sold, it is neither loss or failure for SpeedMass. This also indicates that by selling the 200000 units of cars the company will get some money back, and profit will take place.

Therefore, for each product sold, some profit (according to the profit margin) can get back to the company. However, it is important to note that all costs, as well as the selling price, are based on estimates. Consequently, it is not possible to determine with certainty what the actual profit will be. While a profit margin of a slight amount per unit appears reassuring and promising at first glance, there are several significant risks that must be taken into consideration. Since the calculations rely on assumptions, any underestimation of costs, such as labor and overhead, could result in a lower actual profit per unit or even lead to losses. Furthermore, certain potential costs, such as shipping, marketing, and returns, were not included in the cost estimation. These factors can substantially impact the final cost of SpeedMass products.

Relying on a some profit per unit is inherently risky, as this figure assumes that SpeedMass will sell 200,000 units, a key premise established in the earlier analysis. This sales volume reflects high demand and benefits from economies of scale. However, if demand decreases, it may ultimately incur a financial loss.

2.5.4 Failure check analysis

Failure check is reviewing manufacturing processes in addition to the assumptions that have been made to identify where failures (errors, defects, or unexpected costs) and potential bottlenecks might occur, and planning accordingly. Therefore, a systematic failure check should be conducted in the future to assess potential points of error or risk throughout the production and sales process. One of the important steps that should be taken into consideration during the failure check is *evaluating product quality to identify and minimize defect rates that could lead to returns*. This process can be done by *Income Quality Control (IQC)*. This process is inspecting the raw materials that SpeedMass orders, such as the plastic sheets for the vacuum forming, axels, materials for the mold, etc. They get checked before the production starts by performing random sampling of the ordered patches and setting measurement tools to verify the dimensions and the materials.

Secondly, an additional process should be implemented accordingly, namely *In-Process Quality Control (IPQC)*. This process involves systematic monitoring and inspection during production to enable early detection of defects. For example, during the injection molding process, a defect occurred due to a random error in the Moldex software, which led to incorrect pressure readings. The recorded pressure required to inject the plastic into the mould was lower than the actual pressure required. After recalibration, the machine was able to inject the plastic into the mould correctly; however, some material losses were incurred. Another instance relates to the mould design itself. The moulds should have been fabricated with additional ejection holes to facilitate the removal of the injection-moulded frame from the cavity. As this design consideration was not implemented, the frame became lodged within the mould. The only viable method of extraction involved applying heat to the mould to partially melt the plastic, after which the mould could be reused.

These examples underscore the importance of in-process quality control, as it ensures that all production steps proceed as intended. By continuously monitoring machine displays and verifying that critical parameters, such as injection pressure, are correctly configured, potential defects can be identified at an early stage before they result in material losses.

For the final quality check, it takes place by final inspection on the final product. The inspection on the product can take place in 2 ways:

1. **Manual inspection:** This inspection happens manually through employees from the company, they check the toy cars randomly, not one by one but a random check.
2. **Through an inspection machine:** The inspection machine detects if something is wrong by referencing the product to the main standards that were put for the product.

Both options have an impact on overall costs. Assuming a production volume of only 200000 units, purchasing an inspection machine would not be cost-effective, as the fixed cost per unit would remain relatively high. Economically, such an investment would only be justified if a significantly larger quantity were produced, enabling more efficient distribution of the fixed costs. This approach, however, implies that labour costs will increase proportionally with the number of inspection hours required. When comparing the potential cost escalation associated with both options, the analysis indicates that continuing with manual inspection performed by employees is the more financially viable solution under the current production assumptions. However, if SpeedMass continues manufacturing the same product while maintaining the current profit margin or achieving higher margins, acquiring an inspection machine could become a viable option in the future. Consequently, this consideration leaves SpeedMass with the alternative of manual.

2.6 Conclusion

The process of designing the production layout and the cost analysis is marked by a paradox: in the early stages, there is maximum freedom to make decisions, yet minimal information to assess whether those decisions are optimal. Conversely, by the end of the process, when sufficient information is available to evaluate the quality of earlier choices (like the information that was able to be got from FlexSim, SolidWorks, and general market research), most decisions are already fixed and no longer open to change. This tension between the flexibility and knowledge is evident throughout this project.

Labour costs in the financial analysis were calculated in decimal hours, which does not reflect real-world payment systems where workers are compensated in full hour increments. However, in both the simulation and reality, the total labour required to complete production remains the same. Over the entire production cycle, these rounding differences (which is to the fourth decimal) are expected to balance out.

To summarise this section, the cost calculation for the SpeedMass toy car included material costs for parts and machine costs for processes such as injection molding, vacuum forming, and laser cutting. Assembly and packaging were also factored in, resulting in a total production cost of €6.04 per unit. Combination of cost analysis and production simulation confirms that producing 200,000 toy cars within one year is feasible, specifically within 229 working days. Finally, the project has revealed multiple areas for future improvement, both in terms of the production process itself and the design of the toy car relative to manufacturing efficiency.

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A Injection Moulding Calculations

Set Up Cost	75,7445 €
Set up time (hours)	1
Depreciation cost (€/hour)	0,5459
Purchase price of the machine	10.000 €
Estimated lifespan	10
Labour cost (€/hour)	25,1987
Employee cost per hour	25,1987
Tool cost	50
Machine Cost (per product)	0,7663
Material cost	0,5278 €
Plastic weight per product (g)	19,7000
Material cost per product (€/g)	0,0268
Primary Machine Cost	0,1656 €
Filling cost	0,0083 €
Cooling cost	0,1573 €
Secondary Machine Cost	0,0358 €
Overhead costs	0,0371 €
Mould Cost (Total) (CNC Milling)	974 €
Outsourcing 1 mould	487 €
Number of moulds needed	2

Figure 24: Injection moulding calculations from the Excel sheet

Information for Calculations	
Number of employees	1,00
Working hours (per year)	1.832,00
Filling time (seconds)	1,1570
Filling time (hours)	0,0003
Cooling time (seconds)	22
Cooling time (hours)	0,0061
Injection machine cost per hour (€/h)	25,7445
Secondary time (hours) (cleaning, clamping, etc.)	0,0014
Working hours (per year)	1.832
Overhead cost per hour (€/h)	4,7489
Rent cost	2,7293 €
Annual rent	5.000 €
Utilities cost	1,6376 €
Annual utilities	3.000 €
Insurance cost	0,1092 €
Annual insurance of injection molding (%2 machine value)	200 €
Maintenance cost	0,2729 €
Annual maintenance of injection molding (%5 machine value)	500 €

Figure 25: More injection moulding calculations from the Excel sheet

B Vacuum Forming Calculations

Set Up Cost	151,4891
Set up time (hours)	1
Depreciation cost (€/hour)	1,0917
Purchase price of the machine	20.000 €
Estimated lifespan	10
Labour cost (€/hour)	50,3974
Employee cost per hour	25,1987
Tool cost	100 €
Machine Cost (per product)	0,3349 €
Material cost	0,2091 €
Unit price per car	0,2091 €
Price per sheet (Sheet PETG 2)	6,9 €
Primary Machine Cost	0,0429 €
Vacuuming cost	0,0429 €
Secondary Machine Cost	0,0715 €
Overhead costs	0,0114 €
Mould Cost (Total) (3D Printing)	67.000 €
Outsourcing 1 mould	67 €
Number of moulds needed	1000

Figure 26: Vacuum forming calculations from the Excel sheet

Information for Calculations	
Number of employees	2
Working hours (per year)	1832
Vacuuming time (hours)	0,0008
Vacuuming time (seconds)	60
Vacuum forming machine cost per hour (€/h)	51,4891
Secondary time (hours) (cleaning, clamping, etc.)	0,0014
Secondary time (seconds)	5
Working hours (per year)	1832
Overhead cost per hour (€/h)	5,1310 €
Rent cost	2,7293 €
Annual rent	5.000 €
Utilities cost	1,6376 €
Annual utilities	3.000 €
Insurance cost	0,2183 €
Annual insurance of vacuum forming (%2 machine value)	400 €
Maintenance cost	0,5459 €
Annual maintenance of vacuum forming (%5 machine value)	1.000 €

Figure 27: More vacuum forming calculations from the Excel sheet

C Laser Cutting Calculations

Set Up Cost	132,7314
Set up time (hours)	1
Depreciation cost (€/hour)	7,5328
Purchase price of the machine	138.000 €
Estimated lifespan (years)	10
Labour cost (€/hour)	25,1987
Employee cost per hour	25,1987
Tool cost	100 €
Machine Cost (per product)	0,4326 €
Material cost	0,3462 €
Number of sheets used	735
1 sheet cost (€)	94,2
Primary Machine Cost	0,0213 €
Cutting cost	0,0213 €
Secondary Machine Cost	0,0455 €
Overhead costs	0,0197 €

Figure 28: Laser cutting calculations from the Excel sheet

Information for Calculations	
Number of employees	1
Working hours (per year)	1.832
Cutting time (hours)	0,0007
Cut length (mm)	390
Cutting speed (m/min)	10,00
Number of sheets that will be used	735,29
Laser cutting machine cost per hour (€/h)	32,7314 €
Secondary time (hours) (cleaning, clamping, etc.)	0,0014
Secondary time (seconds)	5
Working hours (per year)	1.832
Overhead cost per hour (€/h)	9,6397 €
Rent cost	2,7293 €
Annual rent	5.000 €
Utilities cost	1,6376 €
Annual utilities	3.000 €
Insurance cost	1,5066 €
Annual insurance of injection molding (%2 machine value)	2.760 €
Maintenance cost	3,7664 €
Annual maintenance of injection molding (%5 machine value)	6.900 €

Figure 29: More laser cutting calculations from the Excel sheet

D CNC Router Calculations

Set Up Cost	127,9279
Set up time (hours)	1
Depreciation cost (€/hour)	2,7293
Purchase price of the machine	50.000 €
Estimated lifespan (years)	10
Labour cost (€/hour)	25,1987
Employee cost per hour	25,1987
Tool cost	100 €
Machine Cost (per product)	0,2850 €
Primary Machine Cost	0,1939 €
Cutting cost - Cutting sheets for vacuum forming into smaller squares	0,0776 €
Cutting cost - Trimming the edges of vacuum formed bodies	0,1164 €
Secondary Machine Cost	0,0388 €
Overhead costs	0,0523 €

Figure 30: CNC router calculations from the Excel sheet

Information for Calculations	
Number of employees	1
Working hours (per year)	1.832
Cutting time (hours)	0,0028
Cutting time (seconds)	10
Trimming time (hours)	0,0042
Trimming time (seconds)	15
CNC router machine cost per hour (€/h)	27,9279 €
Secondary time (hours) (cleaning, clamping, etc.)	0,0014
Secondary time (seconds)	5
Working hours (per year)	1.832
Overhead cost per hour (€/h) 6,2773 €	
Rent cost	2,7293 €
Annual rent	5.000 €
Utilities cost	1,6376 €
Annual utilities	3.000 €
Insurance cost	0,5459 €
Annual insurance of injection molding (%2 machine value)	1.000 €
Maintenance cost	1,3646 €
Annual maintenance of injection molding (%5 machine value)	2.500 €

Figure 31: More CNC router calculations from the Excel sheet

E Ordered Parts Material Costs Calculations

Material Name	Price
Wheels	0,2340 €
Per unit	0,0585 €
M4 Shafts	0,0964 €
Per Unit length (1m)	0,5917 €
M5 Nuts	0,0790 €
Per unit	0,0395 €
M5 Bolts	0,0932 €
Per unit	0,0466 €
M5 Washers	0,0394 €
Per unit	0,0197 €
Ejector pins	0,9900 €
Per unit	0,3300 €
Drilled wheels (ordered)	1,26 €
Total (per car)	2,7920 €

Figure 32: Ordered parts calculations from the Excel sheet

Outsourcing Drilling wheels	
time to drill 1 car's wheels (4 wheels) (sec)	180
Total drill time (hours)	10000
Total price for drilling	251.986,90 €
Cost for one car	1,26 €

Figure 33: More ordered parts calculations from the Excel sheet

F Assembly Costs Calculations

Assembly costs	
Workstation	Time per unit (seconds)
WS1	22
WS2	26
WS3	23
WS4	25
Additional stations	
Cutting 2 shafts	10
Trimming edges of vacuum formed sheet	15
Total Time per car (seconds)	121
Time per car (hours)	0,0336
Employee cost per hour(€)	25,1987 €
Employee cost per day (€)	201,5895 €
Number of employees	5
Manhour calculations	
Hours	
Total time needed to assemble 200000 cars (hours)	6722,22
Operation factor (10%)	672,22
Total production time (hours)	7394,44
Breaks (1/8 = 12.5%)	924,31
Total manned time (hour)	8318,75
Total man days (if 1 worker if working)	1040
Total assembly line cost	209.621,6021 €
Assembly cost per car	1,0481 €

Figure 34: Assembly costs calculations from the Excel sheet

G Packaging Costs Calculations

Packaging scenarios	
Insourcing scenario	
Packaging station 1 (sec) > fold the package	20
Packaging station 2 (sec) > glue the package and then close it, with the car inside	10
Total packaging time in seconds	30
Total packaging time in hours	0,0083
Packaging labour cost per day (€)	0,2100 €
Packaging Size (insourcing)	
Car Dimesnions (volume) (mm ³)	553392
Car length (mm)	126
Car width (mm)	72
Car heigth (mm)	61
Package Dimesnions (volume) (mm ³)	956261
Package length (mm)	151
Package width (mm)	86
Package height (mm)	73
Surface area for the package (mm ²) [2(lw+lh+wh)]	60912
Area of chipboard sheet	2976800
Estimate how many boxes per sheet (without layout optimization):	48,8705
Total sheets needed	4166,6667
Price per 1220x2440 sheet (€)	26,9400 €
Total price for the sheets (€)	112.258,9800 €
Packaging material cost per unit (€)	0,5613 €

Figure 35: Packaging costs- Insourced scenario calculations from the Excel sheet

Outsourcing scenario	
Shipping cost per car	0,0351 €
Total shipping cost	7.026,82 €
Packaging shipping cost of 1 parcel	9,17 € according to quicargo shipping website
Amount of parcels needed	766
Boxes that can fit into one parcel	261,4348 1 Parcel size (1m x 0,5m x 0,5m)
Total space needed (m ³)	191,2523
Package size (m ³)	0,0010
Number of packages	200000

Figure 36: Packaging costs- Outsourced scenario calculations from the Excel sheet

H Assembling and Packaging Costs Combined

Inourced packaging+assembly cost per product	1,8194 €
Total inourced packaging+assembly cost	363.878,3987 €
Total outsourced packaging+assembly cost per prod	1,0832 €
Total outsourced packaging+assembly cost	216.648,4220 €

Figure 37: Packaging costs and assembling costs combined from the Excel sheet

I Final Calculations

Injection molding machine set up	75,7445 €
Vacuum forming machine set up	151,4891 €
Laser cutting machine set up	132,7314 €
Toolpath design (CAM)	500 €
FlexSim	500 €
Moldex	500 €
Designer Cost	480 €
Hand saw	20 €
Set Up Costs	2.359,9651 €
Injection Molding Cost	154.234,7059 €
Vacuum Forming Cost	133.982,6651 €
Laser Cutting Cost	86.515,0252 €
Ordered Parts Material Cost	558.395,2329 €
CNC RouterCost	57.008,7336 €
Packaging+Assembly Costs	216.648,4220 €
Total Cost (mass production, with assembly and packaging costs)	1.209.144,7498 €
Cost per product	6,0457 €

Figure 38: Final calculations from the Excel sheet